SRI CHANDRASEKHADRENDRA SARASWATHI VISWA MAHAVIDYALAYA

UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956

ENATHUR, KANCHIPURAM - 631 561 DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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REG NO :11249A399

CLASS : S4

SUBJECT CODE : BCSF244AI2

SUBJECT NAME :DATA SCIENCE TOOLKIT LAB

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BONAFIDE CERTIFICATE

This is to certify that this is the bonafide record of work done by

Mr/Ms. _____V.V.S NIKHIL_____

with Reg. No 11249A399 done by of II-B.E.(CSE) in –

DATA SCIENCE TOOLKIT LAB-BCSF244AI2 for Lab during the academic year 2025 -2026.

Station : ENATHUR

Date :

Staff-in-charge **Head of the Department**

Submitted for the practical examination held on _____

Examine-1 **Examine -2**

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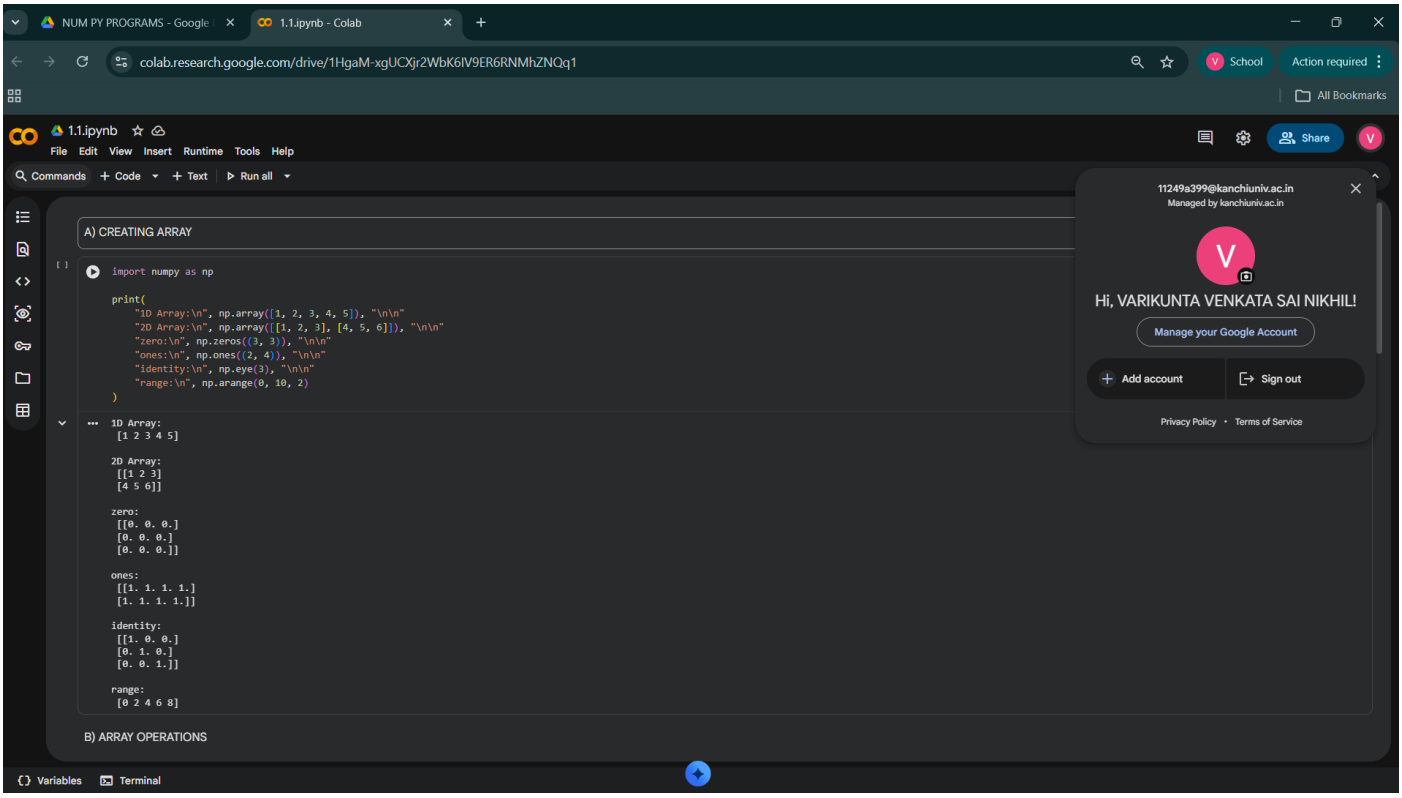
INTRODUCTION TO PYTHON LIBRARIES FOR DATA SCIENCE

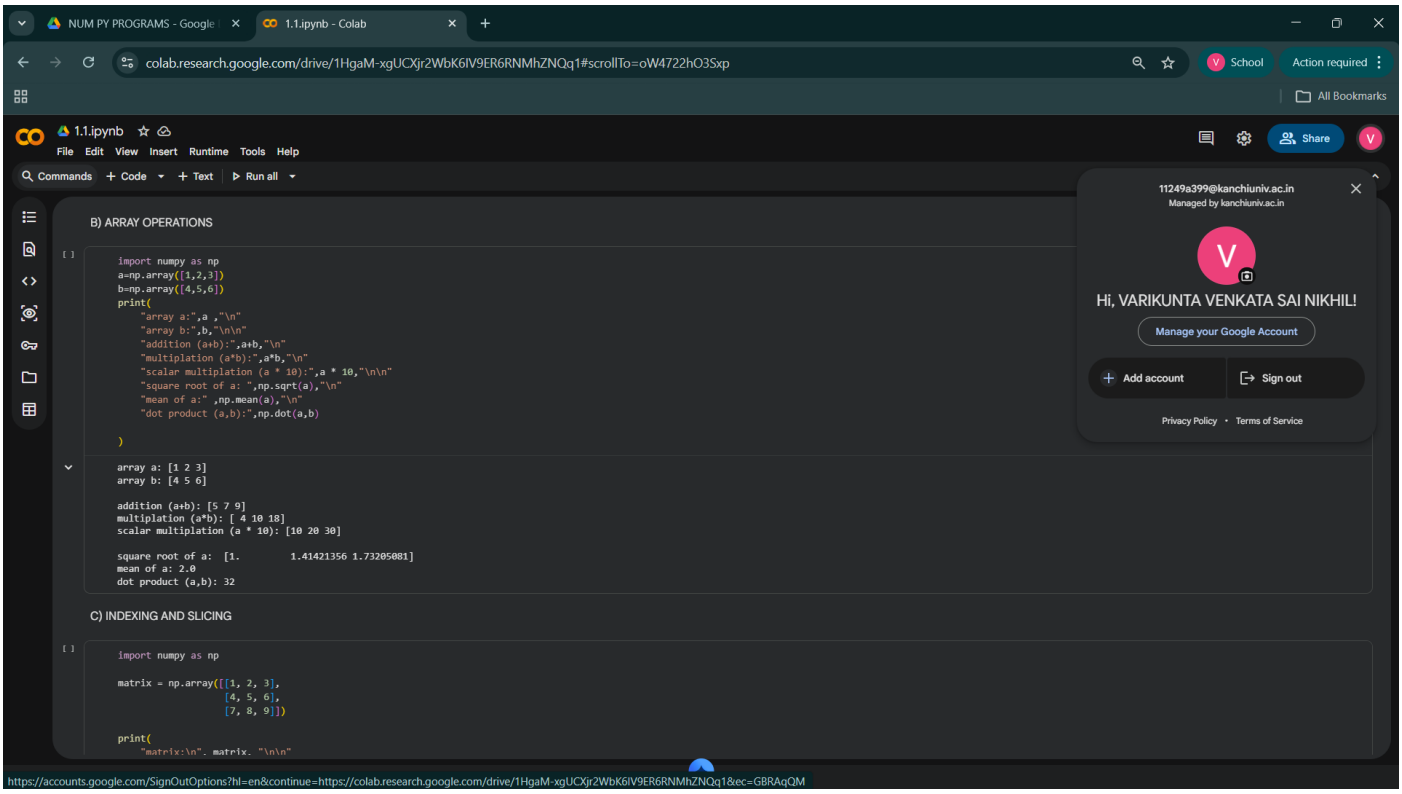
Exp-01(1.1)

Date-02-02-2026

Aim:

To learn basic Python data science libraries like NumPy and Pandas.





NUM PY PROGRAMS - Google | 1.1.ipynb - Colab

colab.research.google.com/drive/1HgaM-xgUCXjr2WbK6IV9ER6RNMhZNQq1#scrollTo=oW4722hO3Sxp

1.1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
square root of a:  [1.         1.41421356 1.73205081]
mean of a: 2.0
dot product (a,b): 32
```

C) INDEXING AND SLICING

```
import numpy as np

matrix = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])

print(
    "matrix:\n", matrix, "\n\n"
    "element at [1,2]:", matrix[1, 2], "\n\n"
    "first row:", matrix[0, :], "\n\n"
    "top-left 2x2 sub matrix:\n", matrix[0:2, 0:2]
)
```

```
matrix:
 [[1 2 3]
 [4 5 6]
 [7 8 9]]

element at [1,2]: 6

first row: [1 2 3]

top-left 2x2 sub matrix:
 [[1 2]
 [4 5]]
```

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1.1.ipynb

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```
top-left 2x2 sub matrix:
 [[1 2]
 [4 5]]
```

*** PANDAS ***

A) CREATING DATAFRAME

```
import pandas as pd
df=pd.read_csv('/content/googlespreadsheet1 - Sheet1.csv')
print(df)
```

```
      NAME REG NUM   AGE SECTION DEPARTMENT
0      NaN     NaN   NaN     NaN        NaN
1     RAJU    A001  20.0       A        CSE
2      SAI    A002  19.0       B        CSE
3  BALAJEE    A003  21.0       C        CSE
4     ROSE    A004  18.0       D        CSE
5     SONU    A005  20.0       E        CSE
```

```
import pandas as pd
df=pd.read_csv('/content/goog spreadsheet2 - Sheet1 (1).csv')
print(df)
```

```
      NAME  MARKS
0     RAJU     99
1     SONU     95
2     MIKE     98
3  SWAROOP     97
```

Welcome To Colab - Colab | NUM PY PROGRAMS - Google | 1.1.ipynb - Colab

colab.research.google.com/drive/1HgaM-xgUCXjr2WbK6IV9ER6RNMhZNQq1#scrollTo=7qQhj03MRZJx

1.1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```python
# 5. Statistical Functions
print("\nMean:", np.mean(a))
print("Median:", np.median(a))
print("Standard Deviation:", np.std(a))
print("Variance:", np.var(a))
print("Minimum:", np.min(a))
print("Maximum:", np.max(a))
print("Sum:", np.sum(a))
print("Percentile (50):", np.percentile(a, 50))

# 6. Axis-based operations
print("\nColumn-wise Sum:\n", np.sum(g, axis=0))
print("Row-wise Mean:\n", np.mean(g, axis=1))

# 7. Indexing & Slicing
print("\nFirst element:", a[0])
print("Slice:", a[1:4])
print("2D Slice:\n", g[0:2, 1:3])

# 8. Boolean & Conditional
print("\nWhere > 50:\n", np.where(g > 50))
print("Any > 90:", np.any(g > 90))
print("All > 0:", np.all(g > 0))

# 9. Sorting & Searching
print("\nSorted a:", np.sort(a))
print("Index of max:", np.argmax(a))
print("Index of min:", np.argmin(a))
print("Unique elements:", np.unique([1, 2, 2, 3, 4, 4]))

# 10. Linear Algebra
x = np.array([[1, 2], [3, 4]])
y = np.array([[5, 6], [7, 8]])

print("\nDot Product:\n", np.dot(x, y))
print("Matrix Multiplication:\n", np.matmul(x, y))
print("Determinant:", np.linalg.det(x))
print("Inverse:\n", np.linalg.inv(x))

# 11. Random Functions
np.random.seed(1)
rand_sample = np.random.choice(a, size=3)
print("\nRandom Choice:", rand_sample)
```

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Variables Terminal

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1.1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

EXP 1.1

```python
import numpy as np

# 1. Array Creation
a = np.array([1, 2, 3, 4, 5])
b = np.arange(1, 10, 2)
c = np.linspace(0, 1, 5)
d = np.zeros((2, 3))
e = np.ones((2, 2))
f = np.eye(3)
g = np.random.randint(1, 100, size=(3, 3))

print("Array a:", a)
print("Arange b:", b)
print("Linspace c:", c)
print("Zeros d:\n", d)
print("Ones e:\n", e)
print("Identity f:\n", f)
print("Random g:\n", g)

# 2. Array Properties
print("\nShape:", g.shape)
print("Dimensions:", g.ndim)
print("Size:", g.size)
print("Data type:", g.dtype)

# 3. Reshape & Transpose
reshaped = g.reshape(1, 9)
print("\nReshaped array:", reshaped)
print("Transpose:\n", g.T)
print("Flattened:", g.flatten())

# 4. Mathematical Operations
print("\nAdd:", np.add(a, 2))
print("Multiply:", np.multiply(a, 3))
print("Square Root:", np.sqrt(a))
print("Power:", np.power(a, 2))
print("Absolute:", np.abs([-1, -2, 3]))
```

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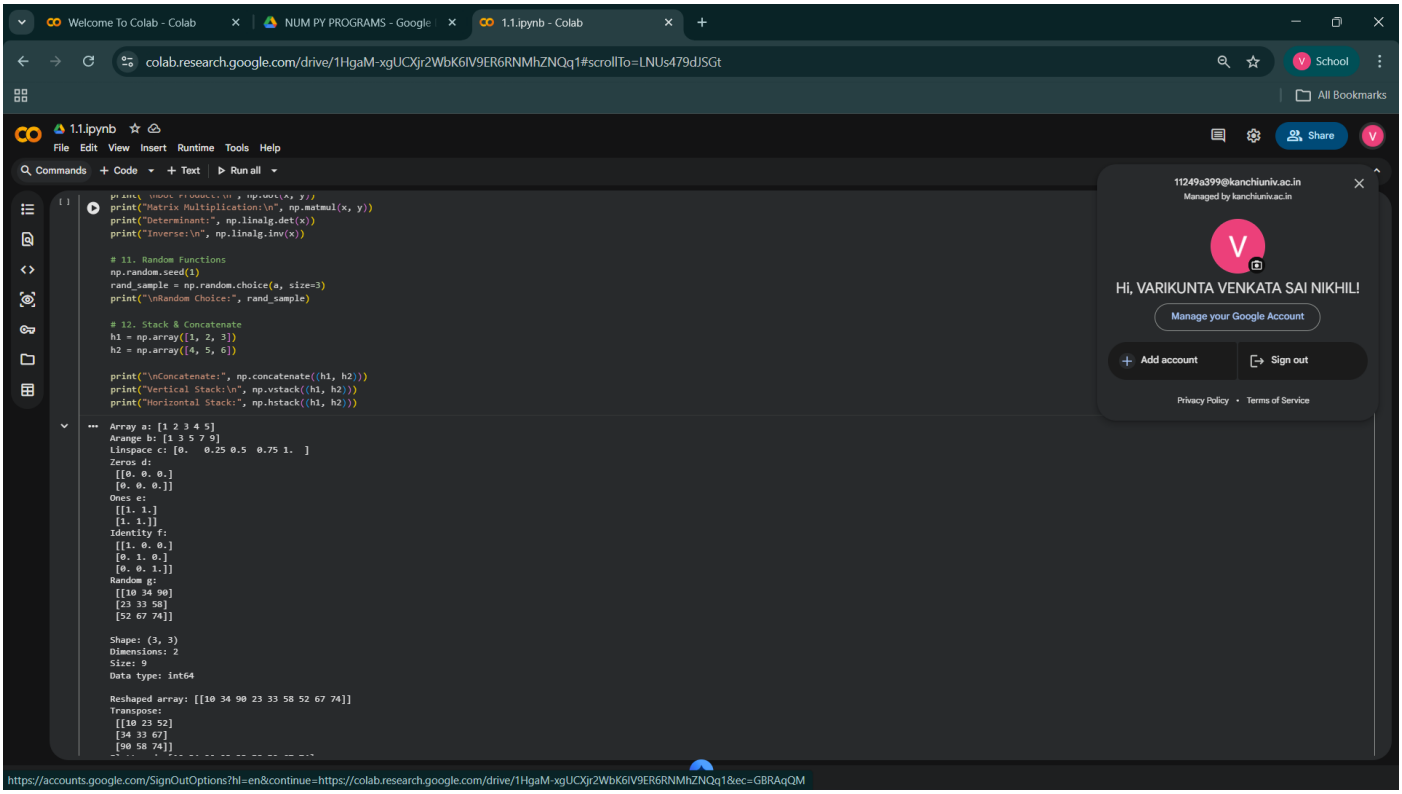
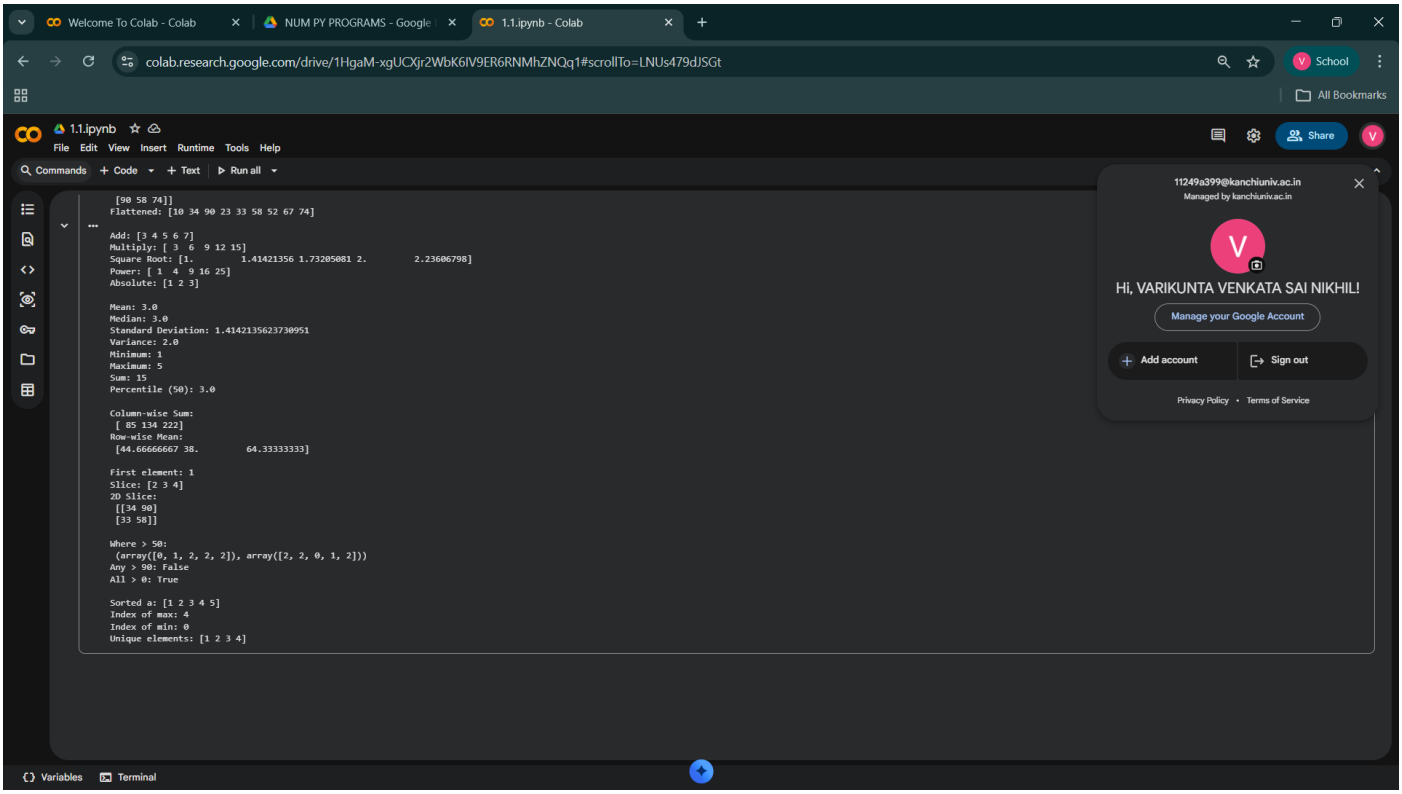
OUTPUT-





Result:

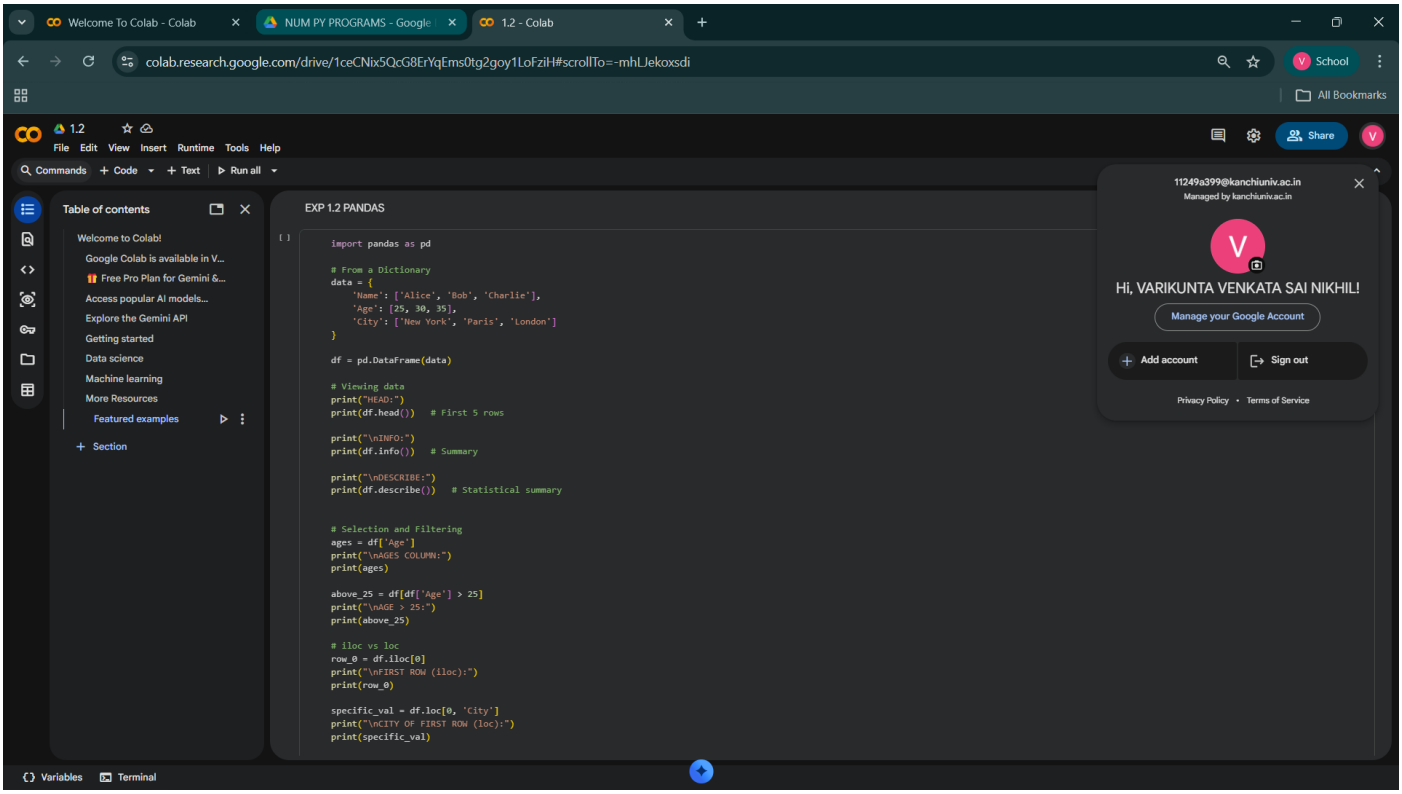
Thus, it is implemented *Introduction to Python Libraries for Data Science* successfully.

PANDAS

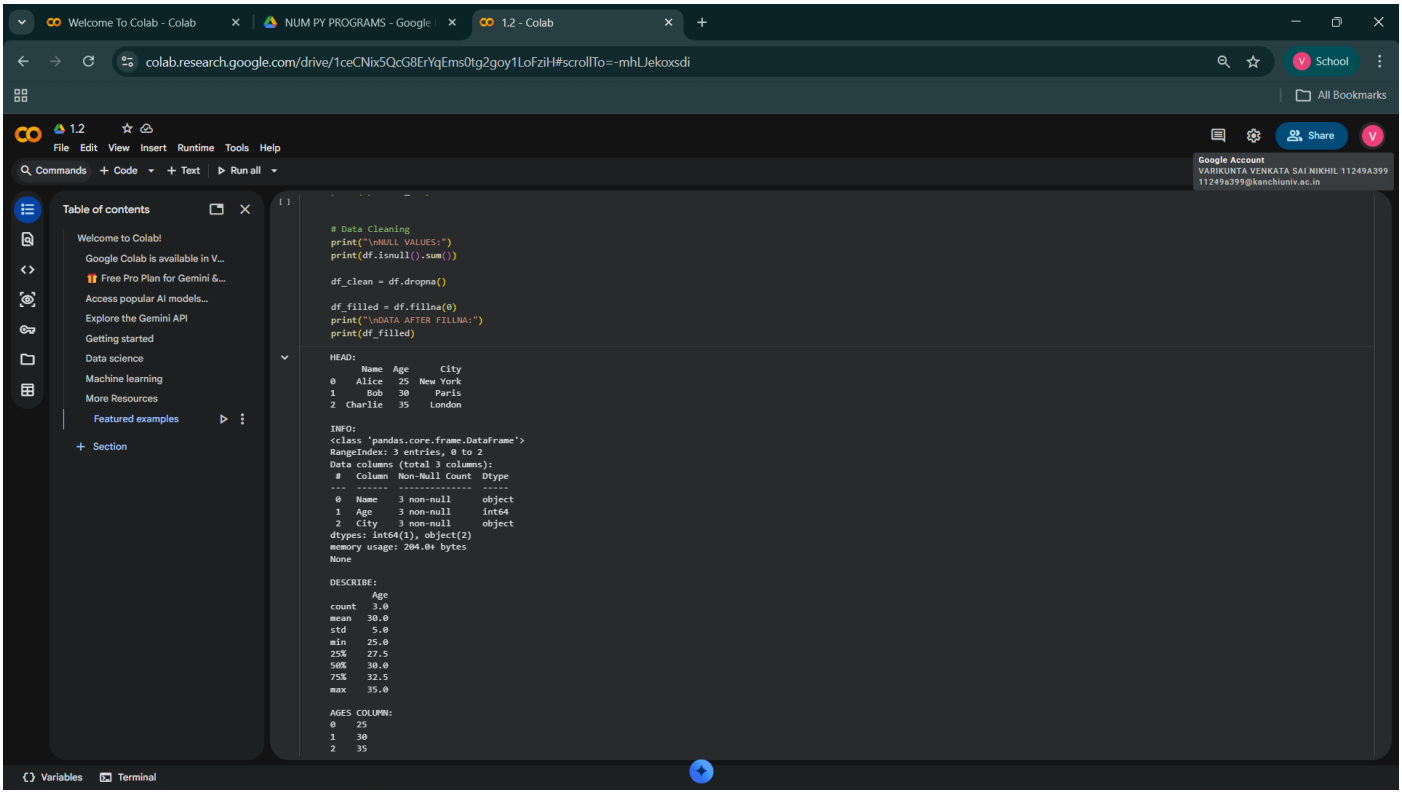
EXP-1.2

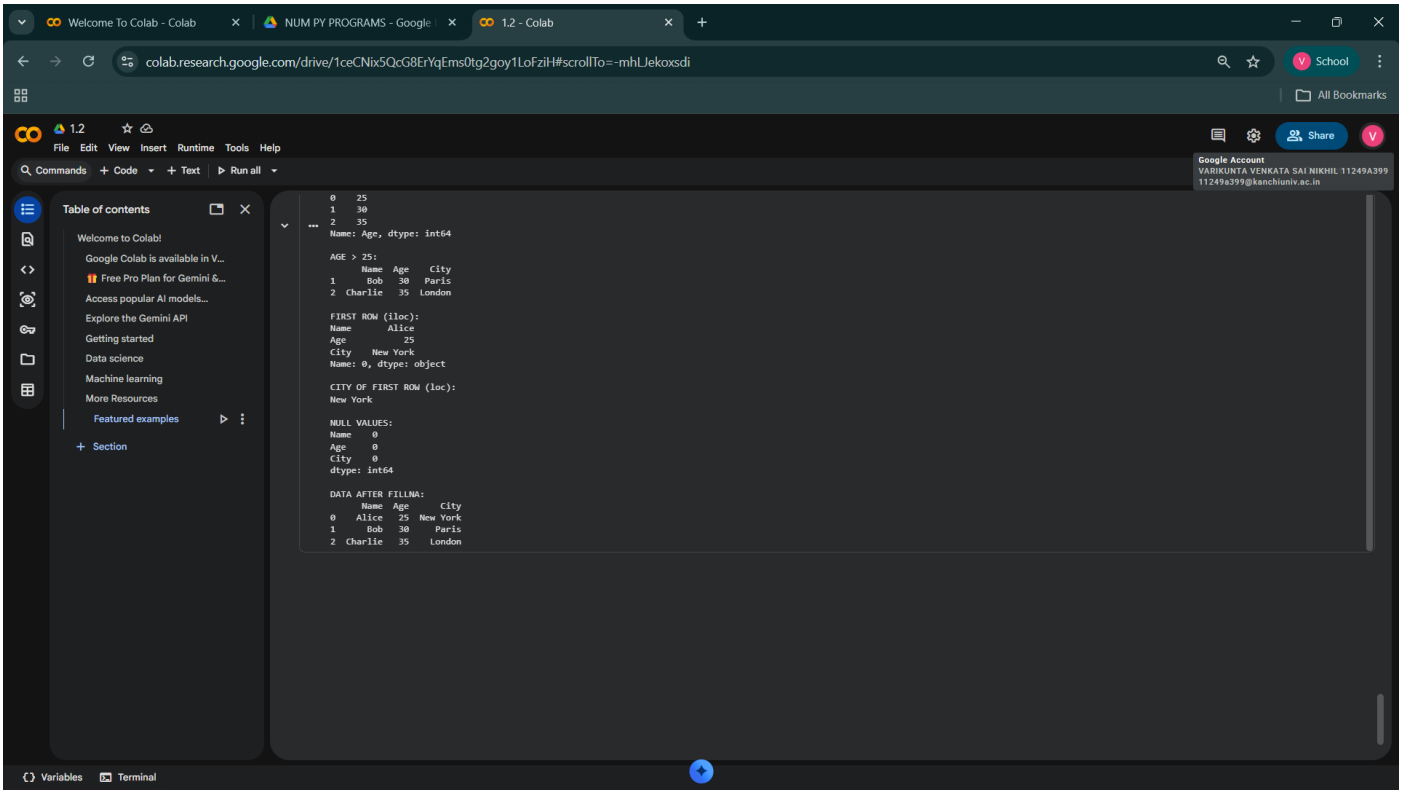
DATE-02-02-2026

Aim:
To study the Pandas library for data handling and analysis.



OUTPUT-





Result:

Thus, it is implemented *Pandas* successfully.

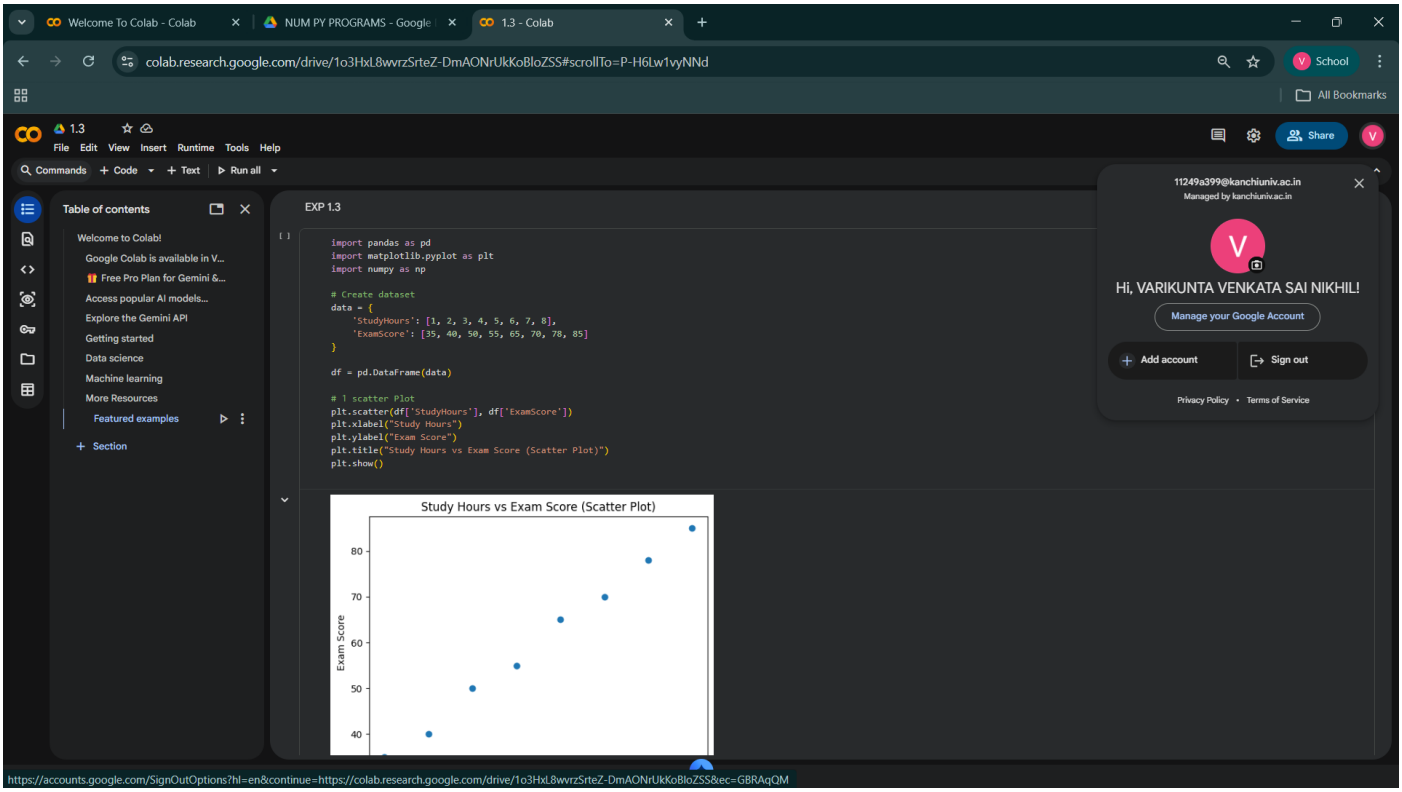
MATPLOT LIB

EXP-1.3 DATE-02-02-2026

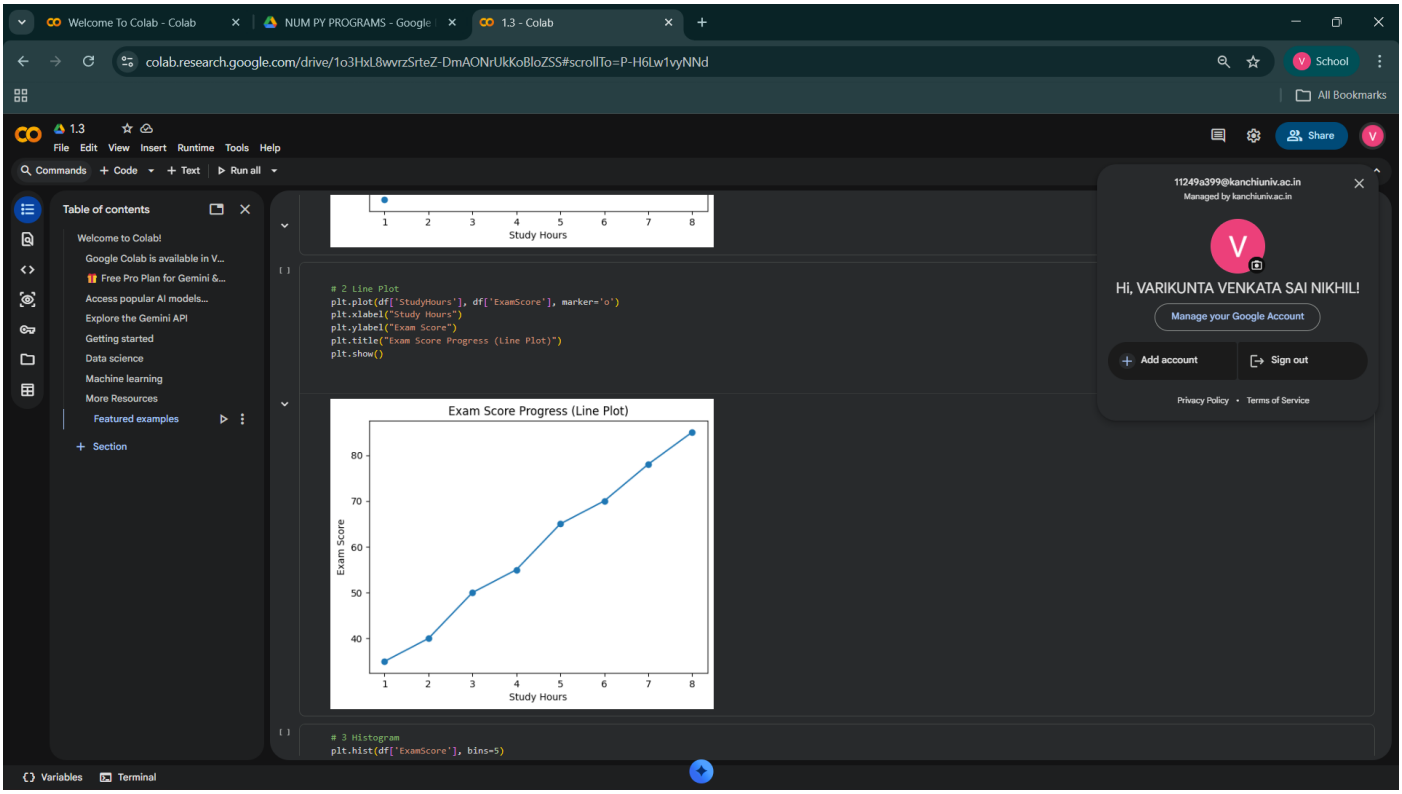
Aim:

To study the Matplotlib library for data visualization.

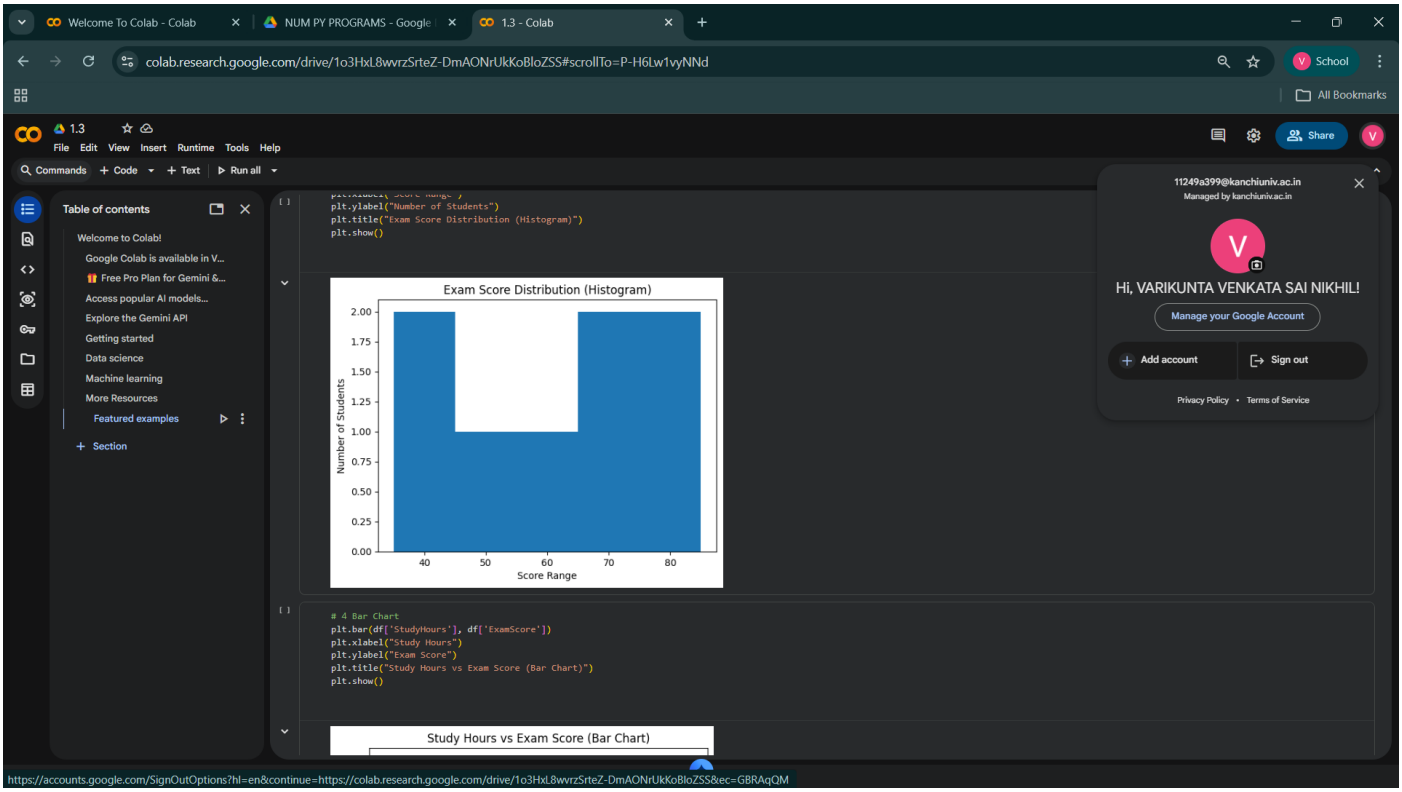
PROGRAM WITH OUTPUT-



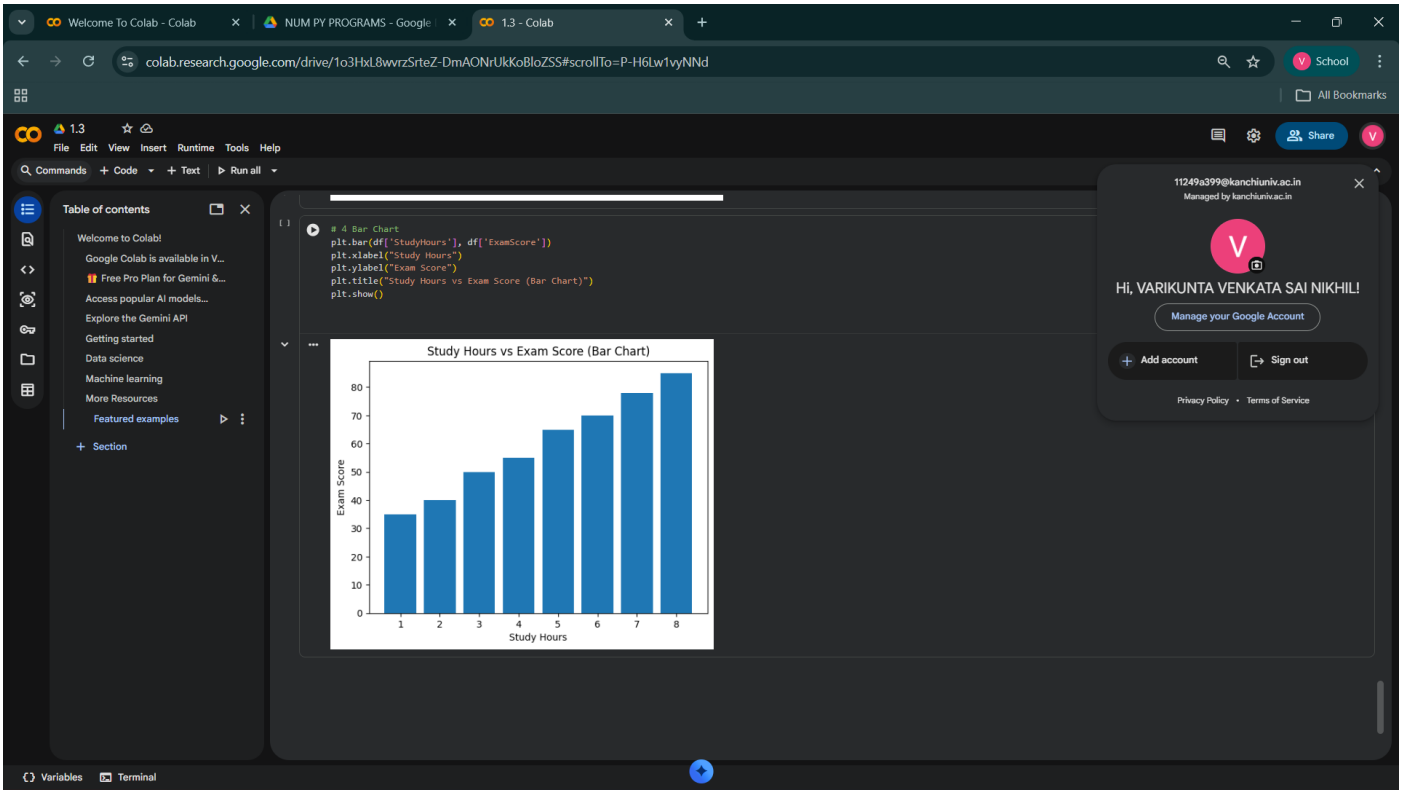
PROGRAM WITH OUTPUT-



PROGRAM WITH OUTPUT-



PROGRAM WITH OUTPUT-



Result:

Thus, it is implemented *Matplotlib* successfully.

SCIKIT LEARN

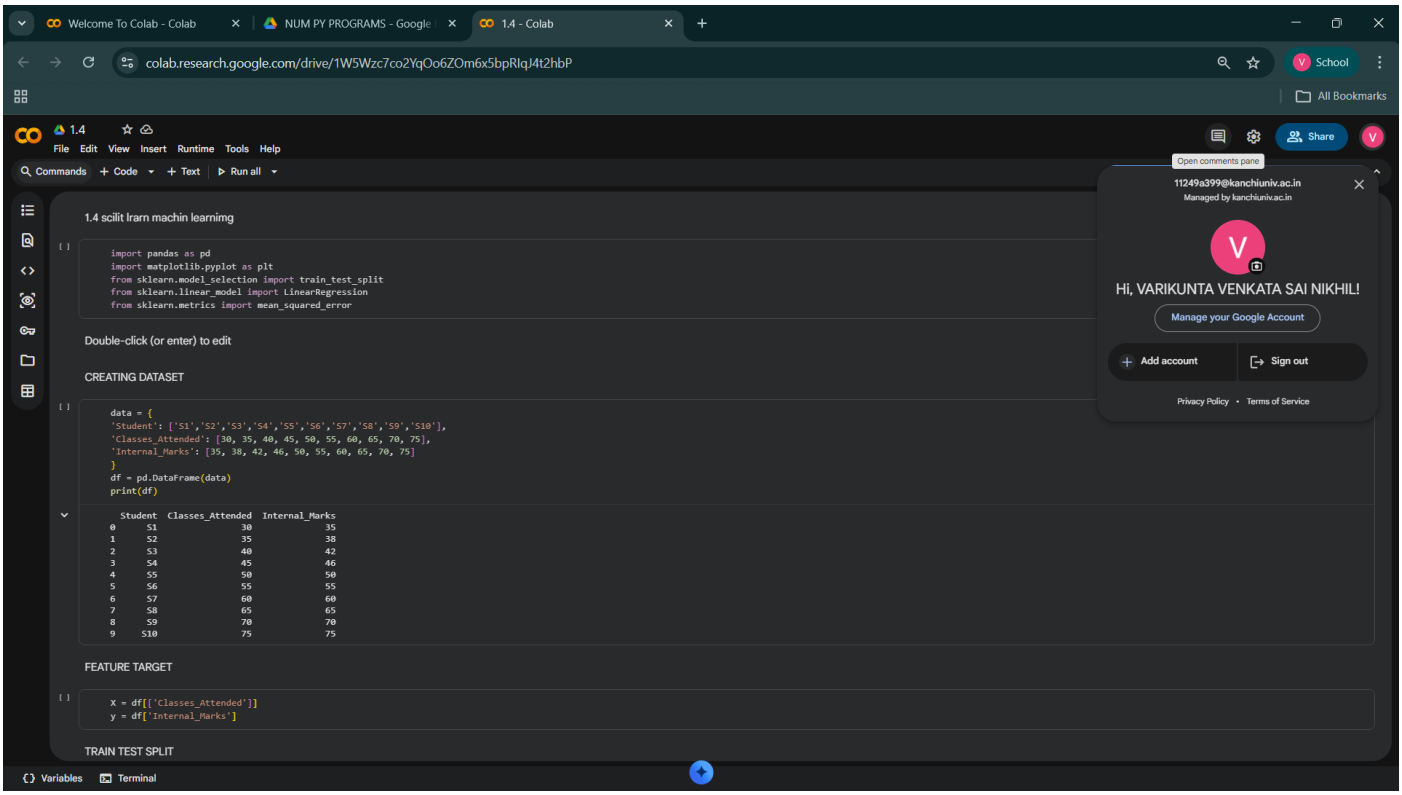
EXP N-1.4

DATE-09-02-2026

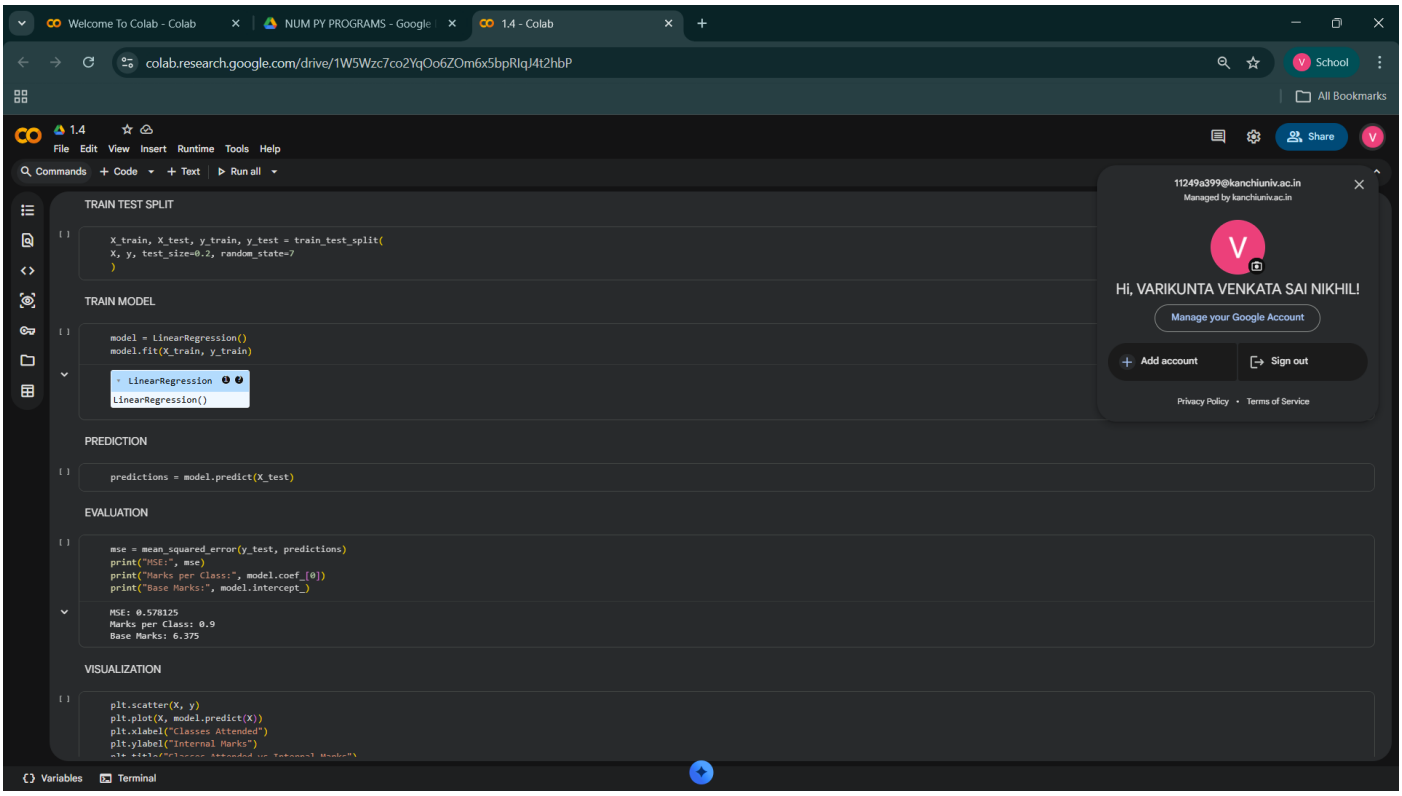
Aim:

To study the scikit-learn library for machine learning.

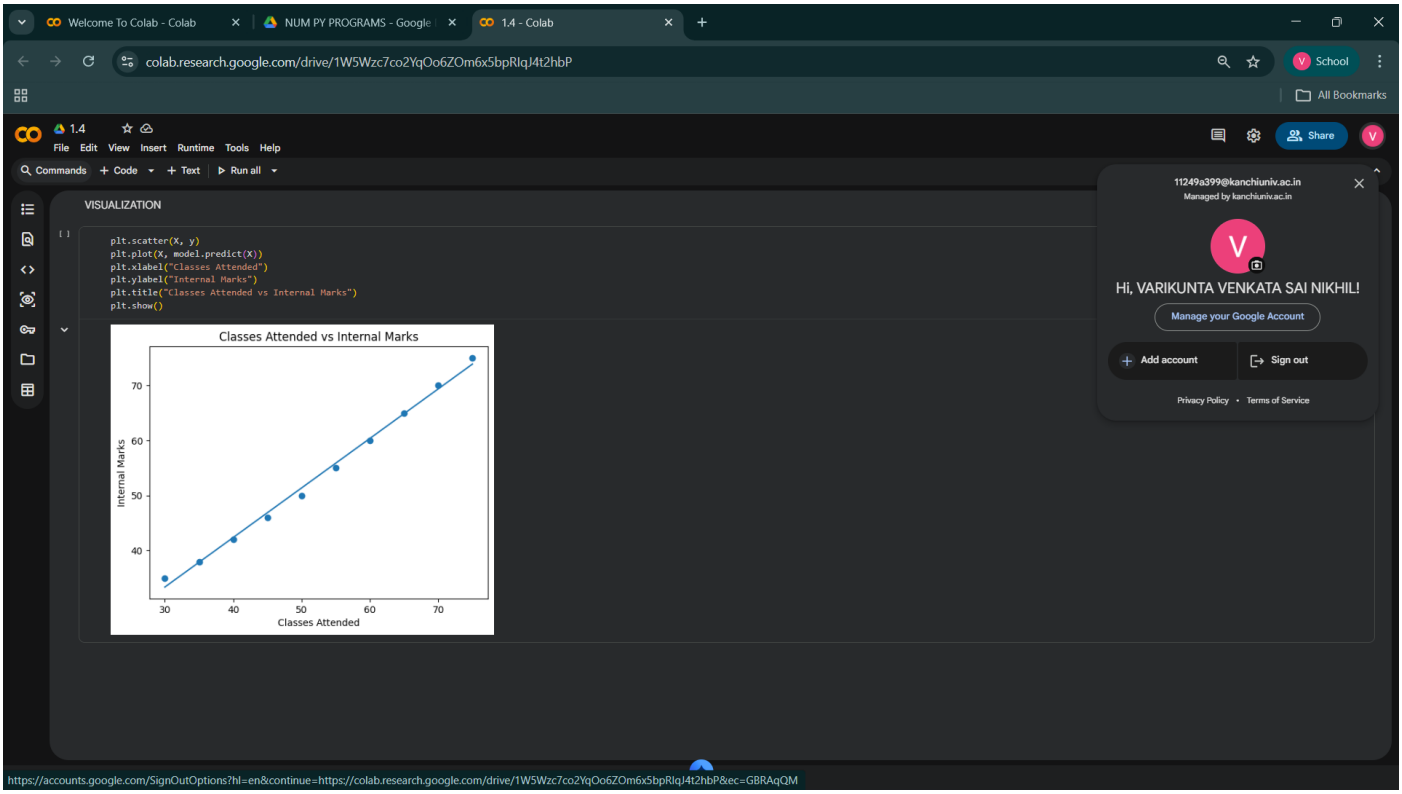
PROGRAM WITH OUTPUT-



PROGRAM WITH OUTPUT-



PROGRAM WITH OUTPUT-



Result:

Thus, it is implemented *Scikit-learn* successfully.

PERFORM DATA EXPLORATION AND PREPROCESSING IN PYTHON

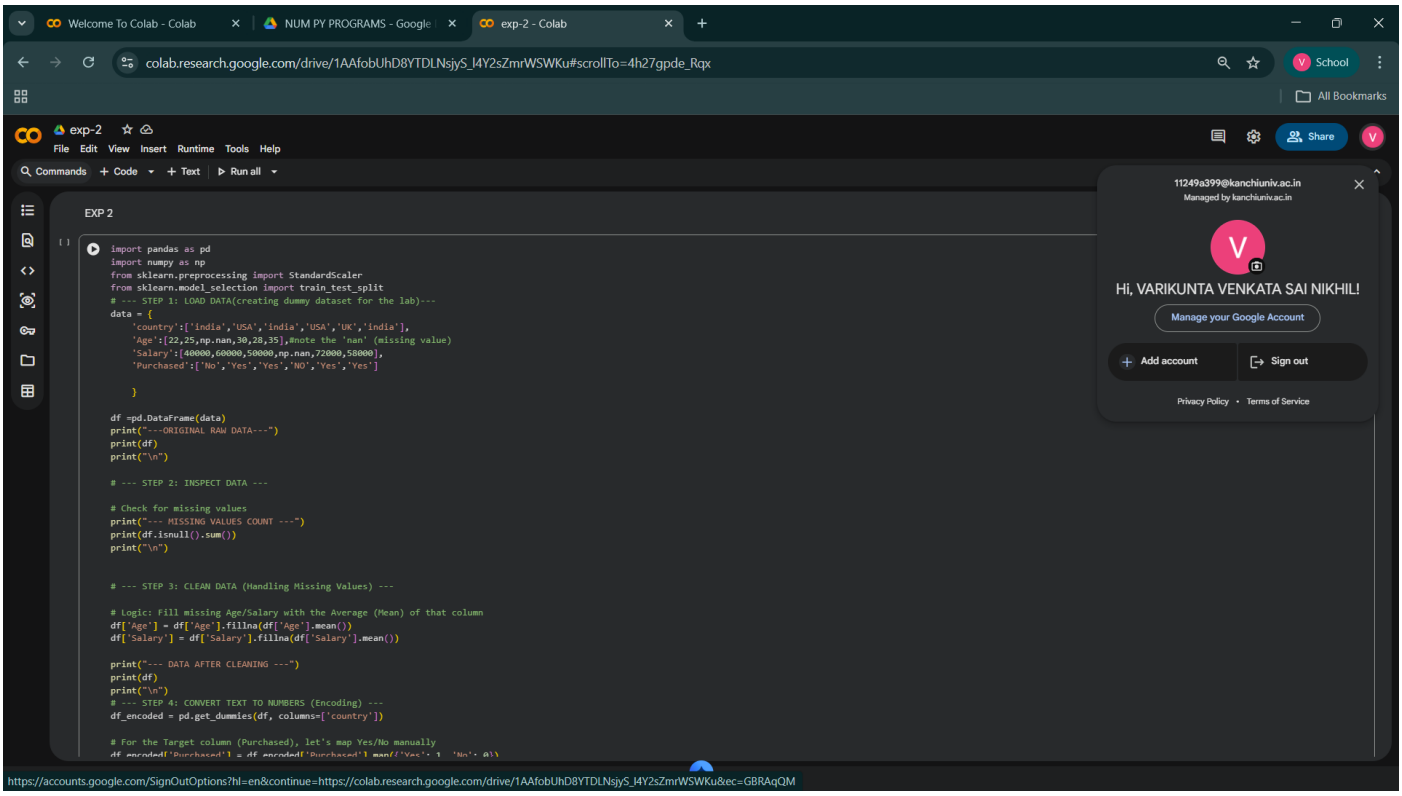
EXP-02

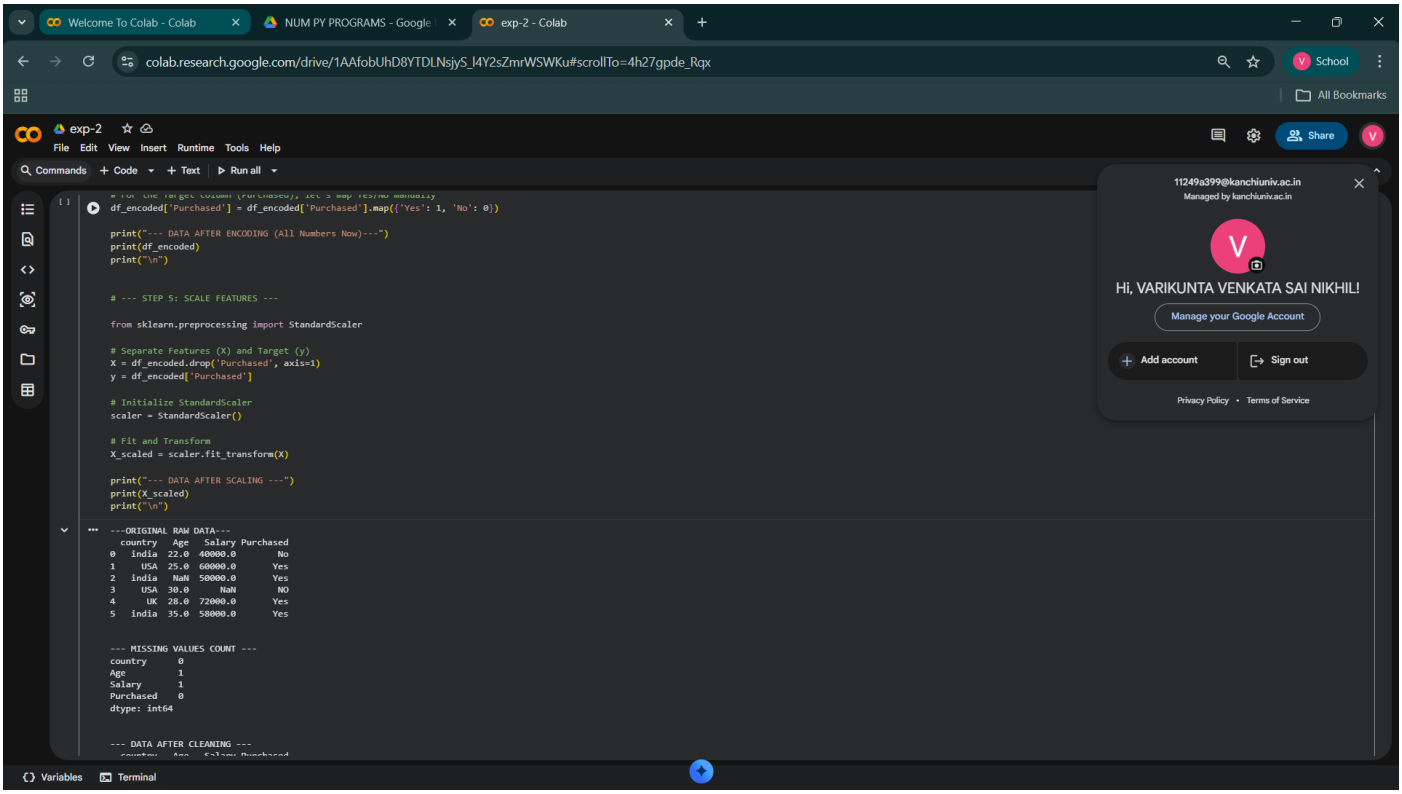
DATE-14-02-2026

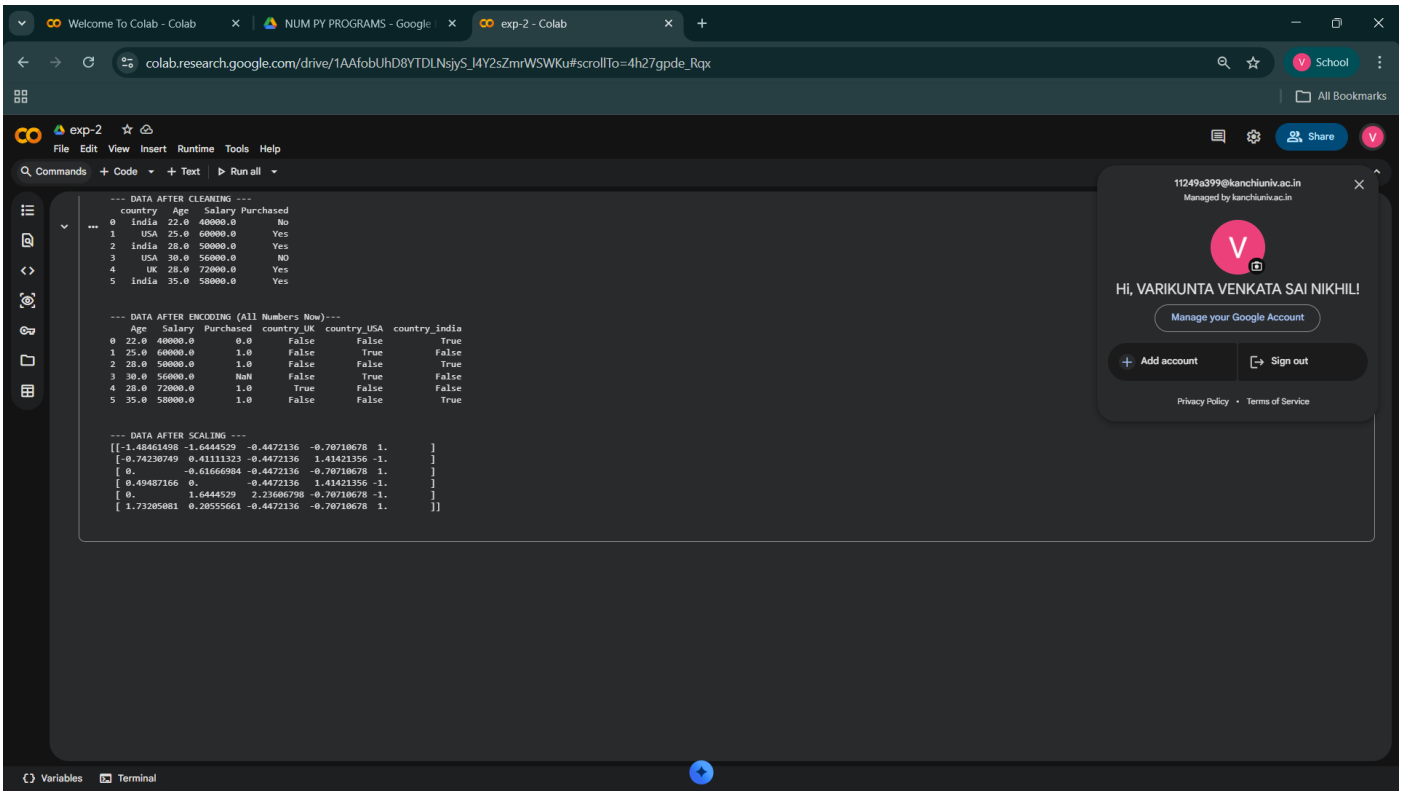
Aim:

To perform data exploration and preprocessing using Python.

PROGRAM WITH OUTPUT-







Result:

The dataset was explored, cleaned, and pre processed successfully, making it ready for further analysis or model building.

IMPLEMENT REGULARIZED LINEAR REGRESSION

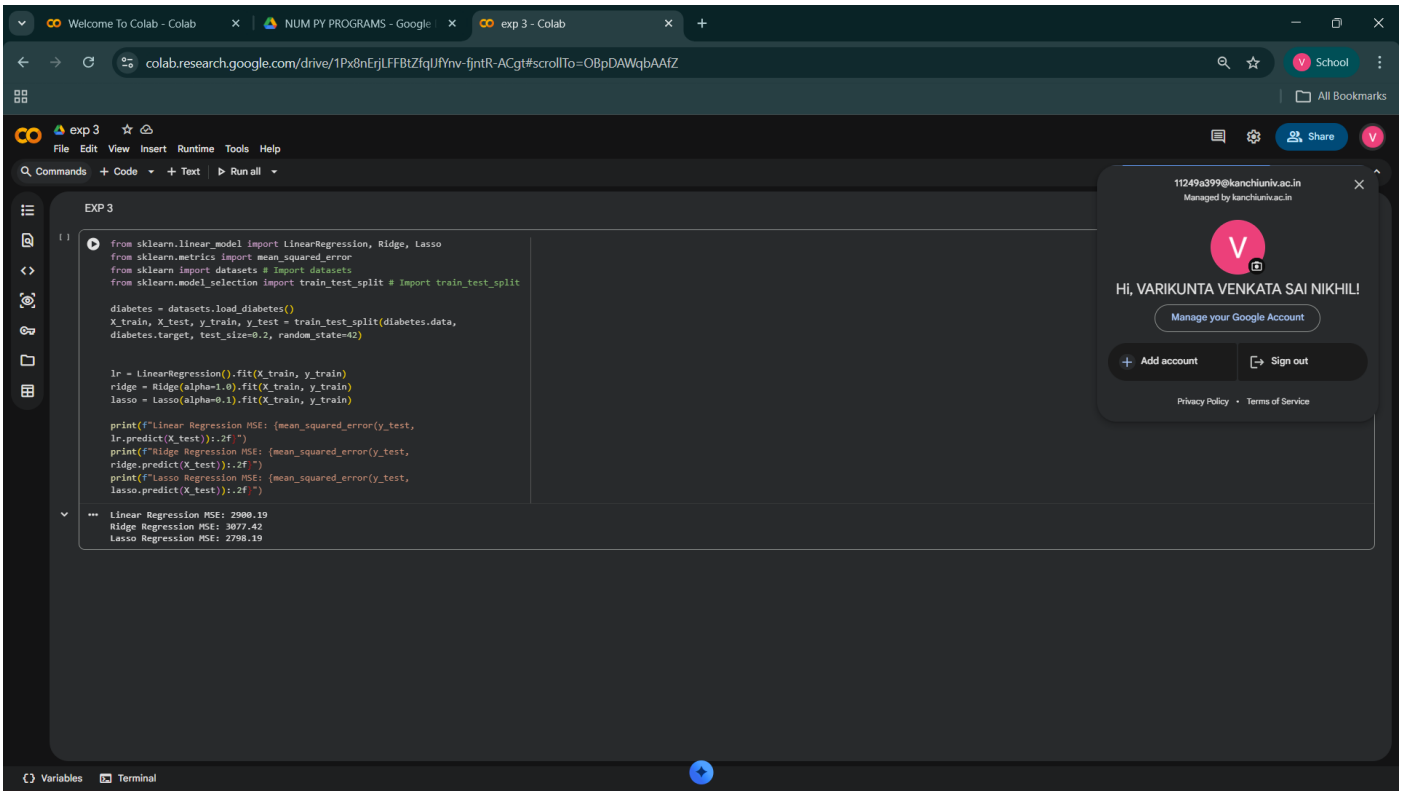
EXP-03

DATE-14-02-2026

Aim

To implement Regularized Linear Regression using Python.

PROGRAM WITH OUTPUT



Result

The regularized linear regression model was successfully implemented using Python. The model reduced overfitting and improved prediction performance by applying regularization techniques such as Ridge or Lasso regression.

IMPLEMENT NAVIE BAYES CLASSIFIER

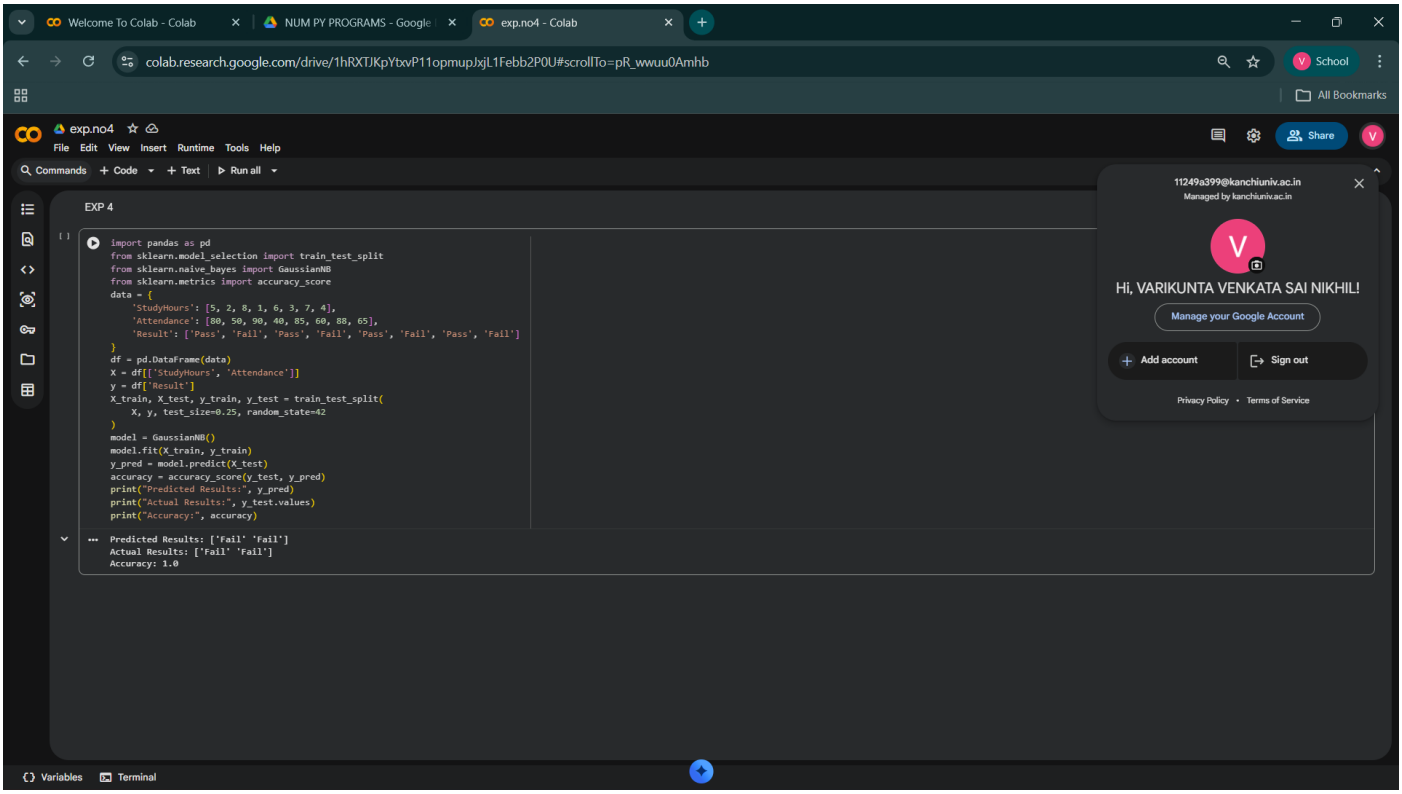
EXP-04

DATE-23-02-2026

Aim

To implement a Naive Bayes classifier using Python for classification of data.

PROGRAM WITH OUTPUT



Result

The Naive Bayes classifier was successfully implemented using Python. The model was trained and used to classify the data effectively, providing accurate predictions based on probability.

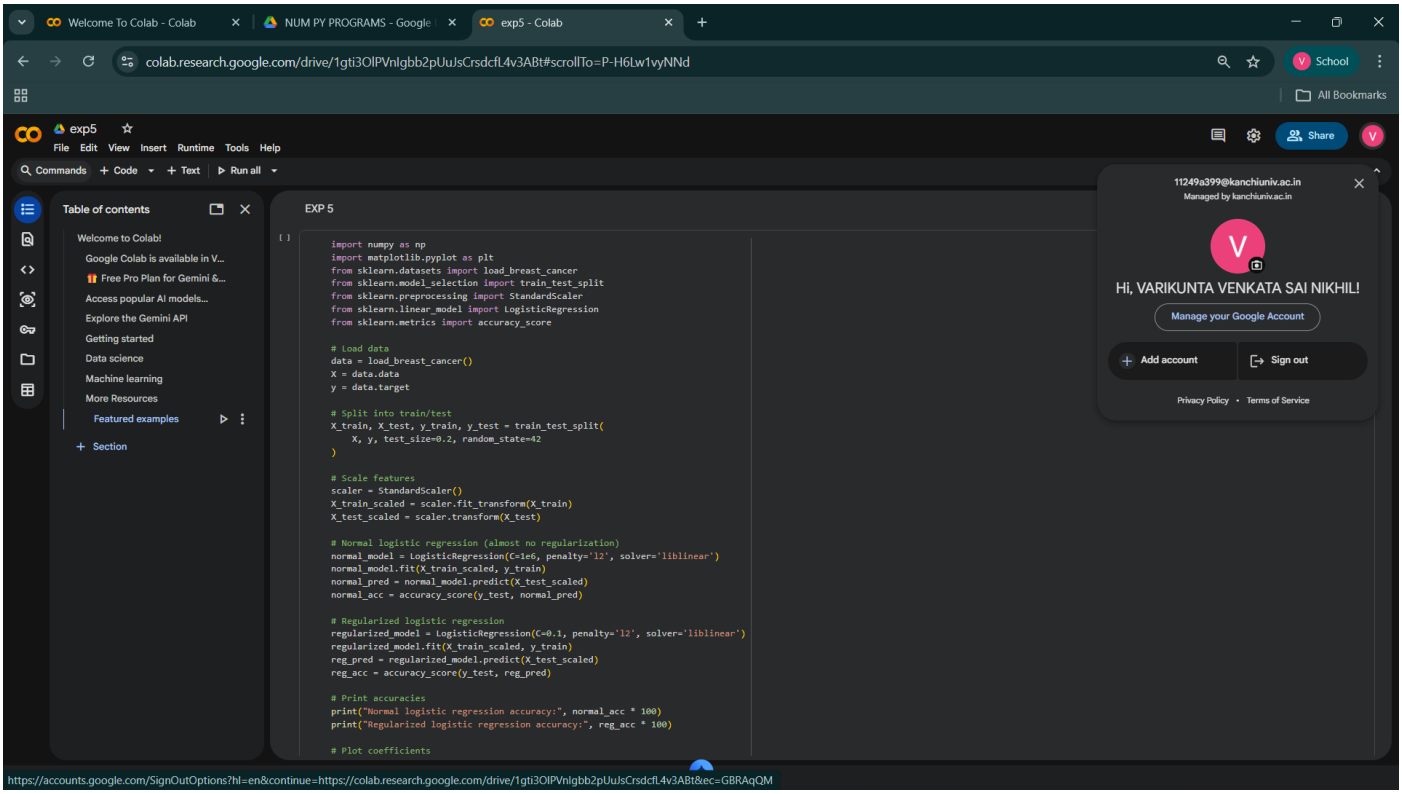
IMPLEMENT REGULARIZED LOGISTIC REGRESSION

EXP-05

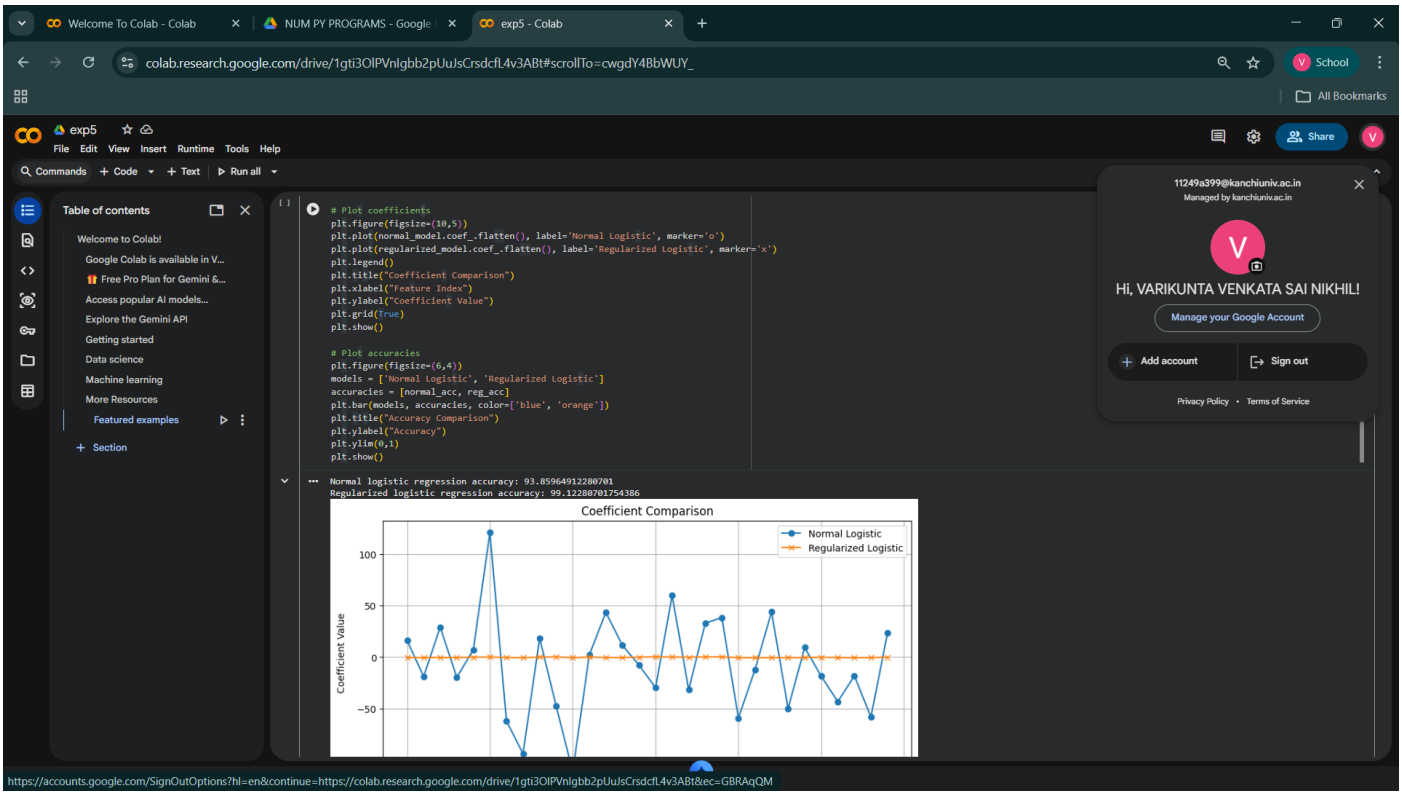
DATE-02-03-2026

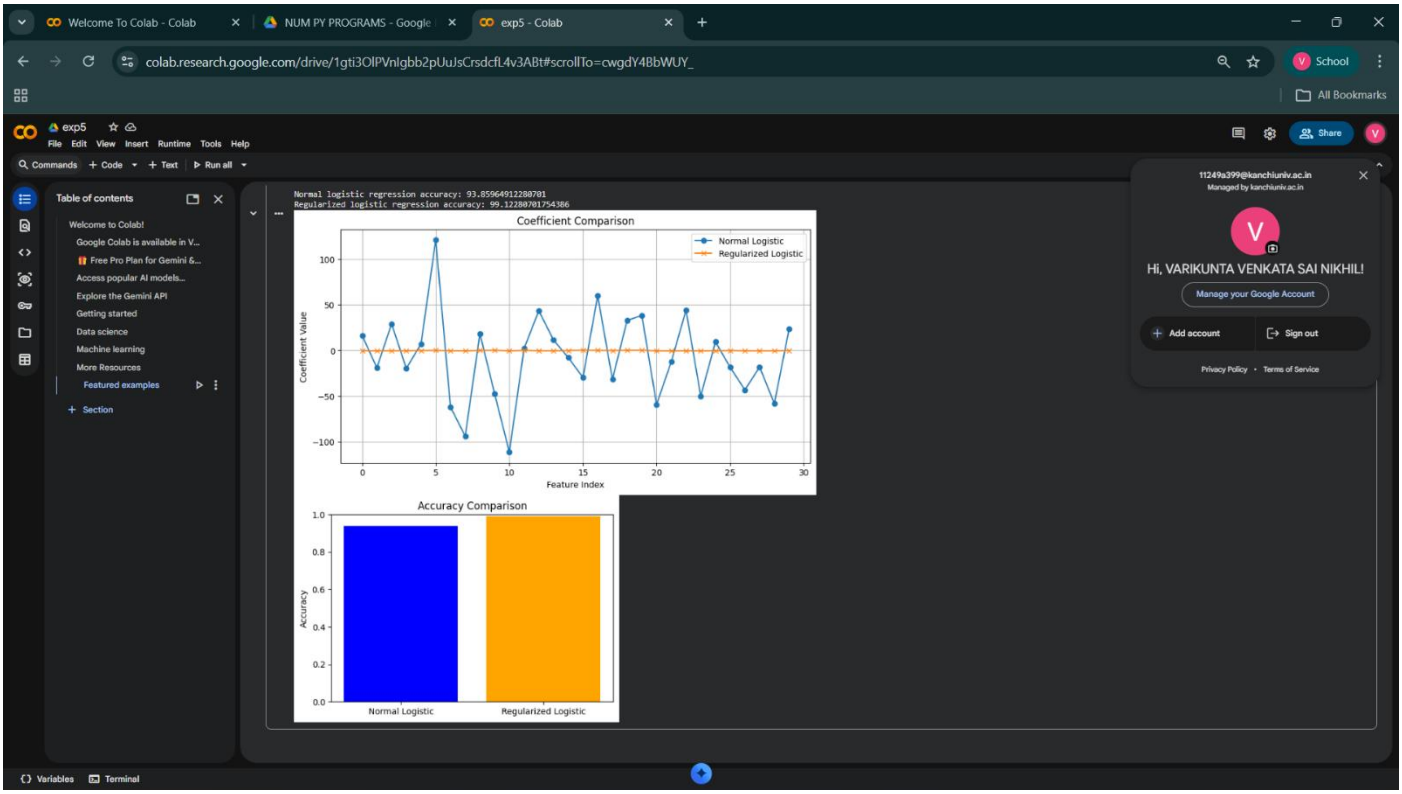
Aim:

To implement Regularized Logistic Regression using Python.



OUTPUT-





Result

The regularized logistic regression model was successfully implemented using Python. Regularization helped reduce overfitting and improved the model's classification performance and generalization on the dataset.

BUILD MODULES USING DIFFERENT ENSEMBLING TECHNIQUES

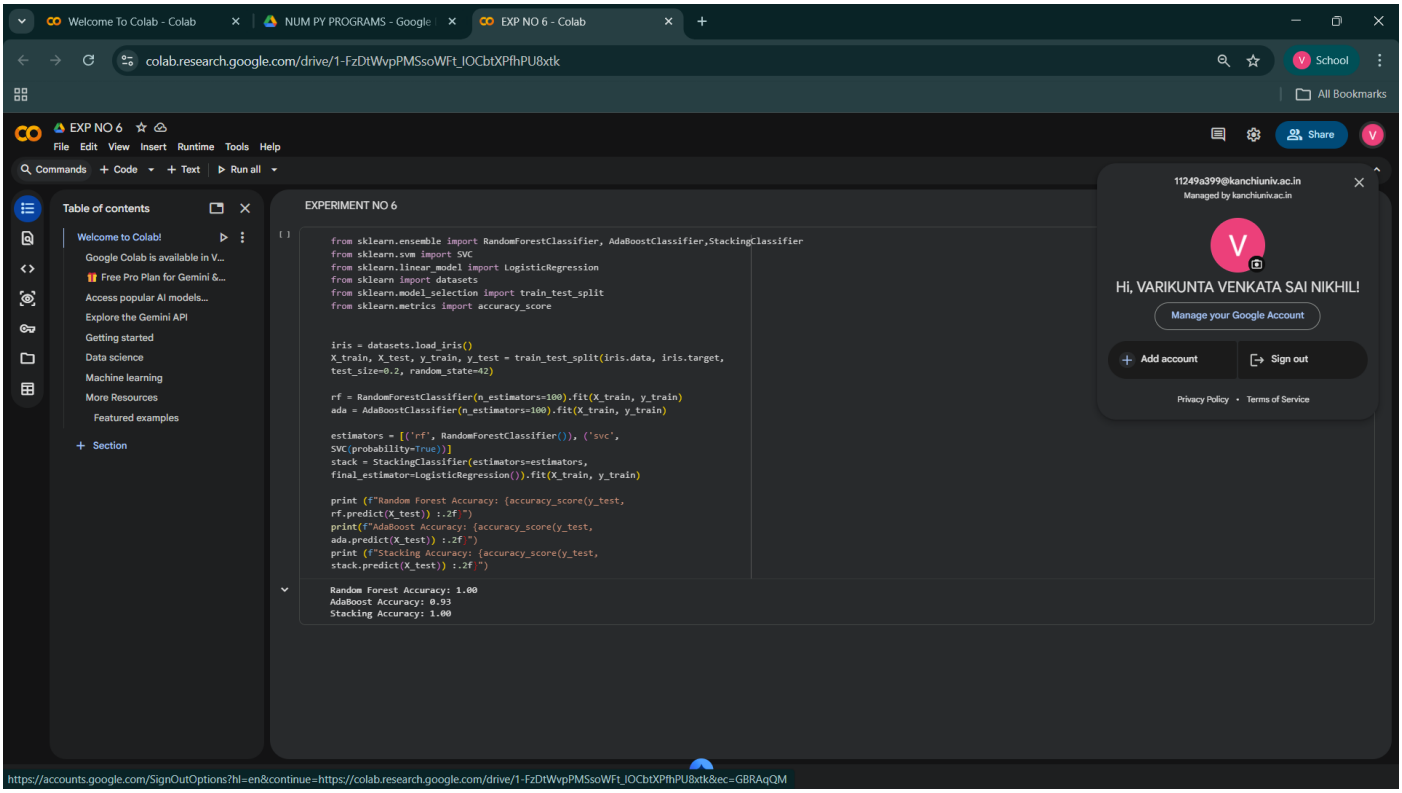
EXP-06

DATE-09-03-2026

Aim:

To build models using different ensemble techniques in Python.

PROGRAM WITH OUTPUT



Result

The ensemble models were successfully built using techniques such as Bagging, Boosting, and Random Forest. These methods improved prediction accuracy and reduced overfitting compared to individual models.

BUILD MODELS USING DECESION TREES

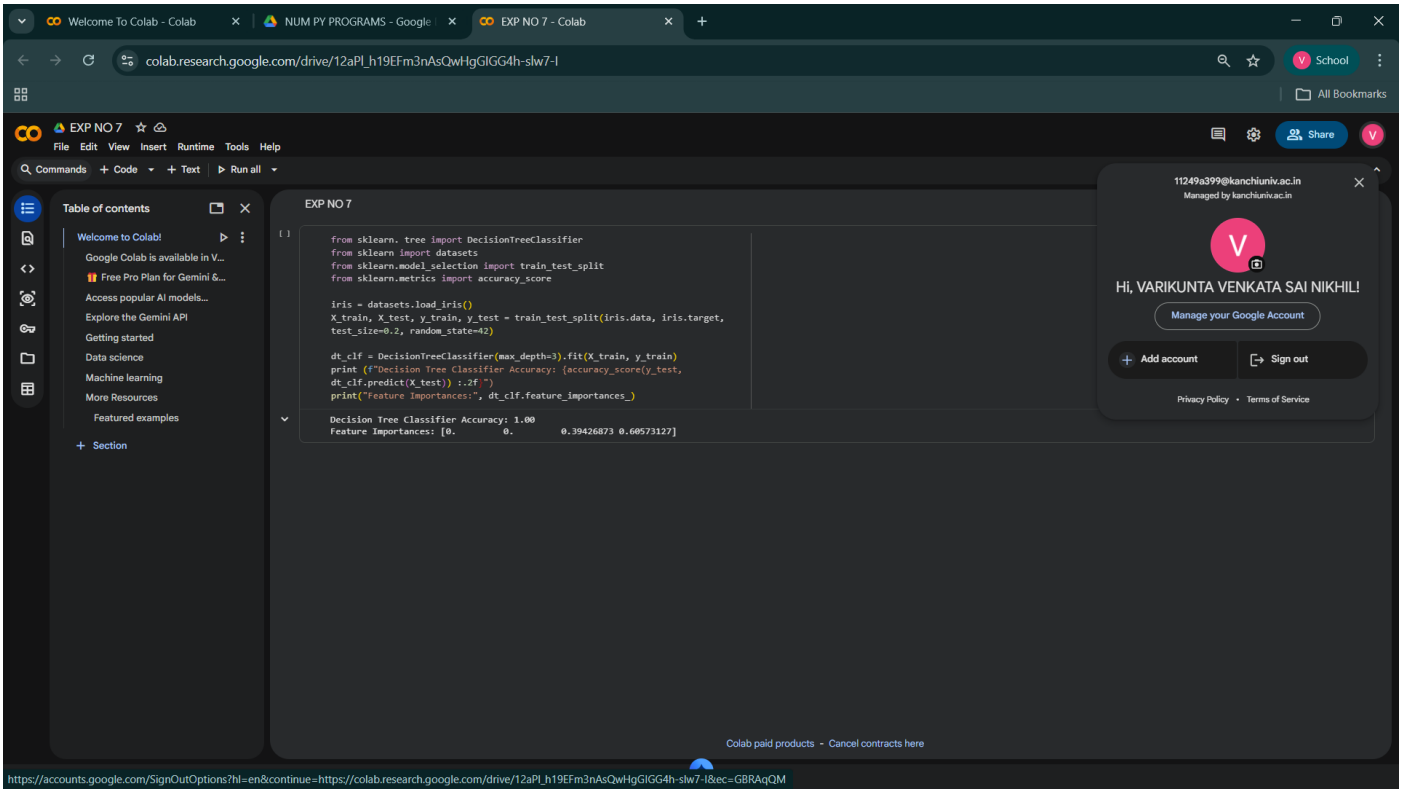
EXP-7

DATE-09-03-2026

Aim:

To build a classification model using Decision Trees in Python.

PROGRAM WITH OUTPUT



Result:

The Decision Tree model was successfully implemented using Python. The model learned decision rules from the dataset and provided accurate predictions for classification tasks.

BUILD THE MODELS USING DIFFERENT KERNAL FUNCTIONS

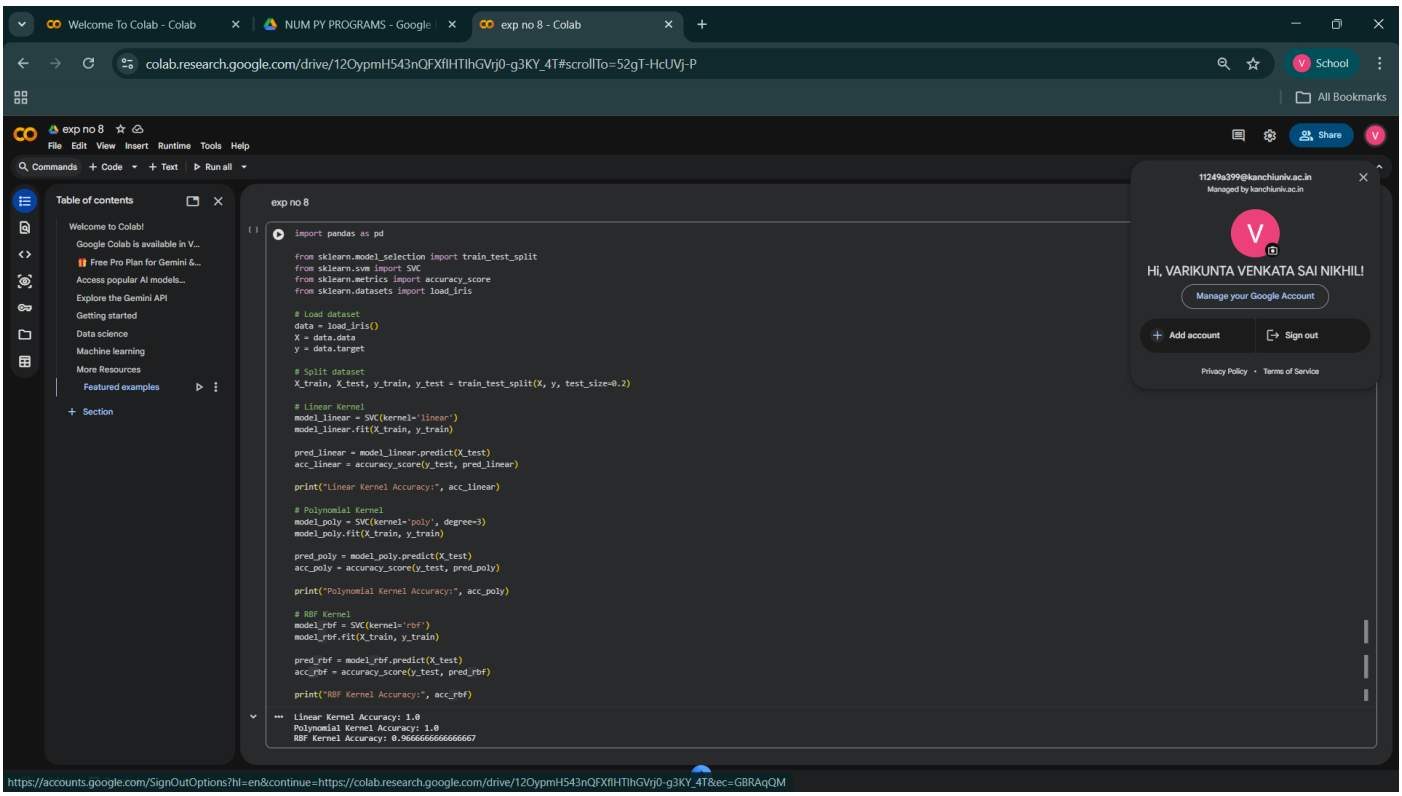
EXP-08

DATE-16-03-2026

Aim

To build models using different kernel functions in Python.

PROGRAM WITH OUTPUT



```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.datasets import load_iris

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Linear Kernel
model_linear = SVC(kernel='linear')
model_linear.fit(X_train, y_train)

pred_linear = model_linear.predict(X_test)
acc_linear = accuracy_score(y_test, pred_linear)

print("Linear Kernel Accuracy:", acc_linear)

# Polynomial Kernel
model_poly = SVC(kernel='poly', degree=3)
model_poly.fit(X_train, y_train)

pred_poly = model_poly.predict(X_test)
acc_poly = accuracy_score(y_test, pred_poly)

print("Polynomial Kernel Accuracy:", acc_poly)

# RBF Kernel
model_rbf = SVC(kernel='rbf')
model_rbf.fit(X_train, y_train)

pred_rbf = model_rbf.predict(X_test)
acc_rbf = accuracy_score(y_test, pred_rbf)

print("RBF Kernel Accuracy:", acc_rbf)
```

```
Linear Kernel Accuracy: 1.0
Polynomial Kernel Accuracy: 1.0
RBF Kernel Accuracy: 0.9666666666666667
```

Result

The model was successfully implemented using kernel-based methods (such as SVM). Different kernel functions like linear, polynomial, and RBF were used, and they improved classification performance based on the dataset structure.

IMPLEMENT K-NEAREST NEIGHBOURS (KNN)CLASSIFICATION

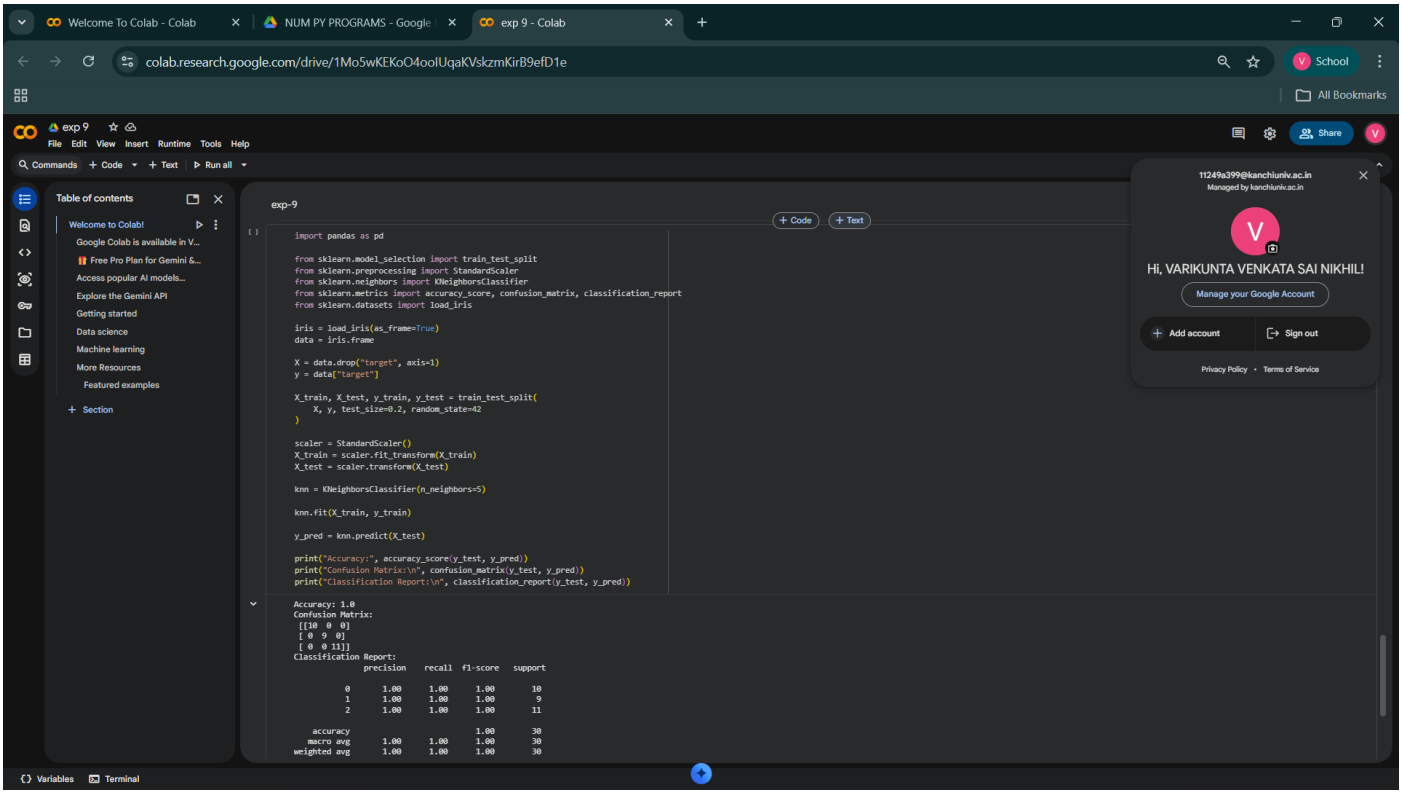
EXP-09

DATE-30-3-2026

Aim:

To implement the K-Nearest Neighbours (KNN) classification algorithm using Python.

PROGRAM WITH OUTPUT



Result

The KNN classification model was successfully implemented using Python. The model classified data based on the nearest neighbors and produced effective and accurate predictions for the given dataset.

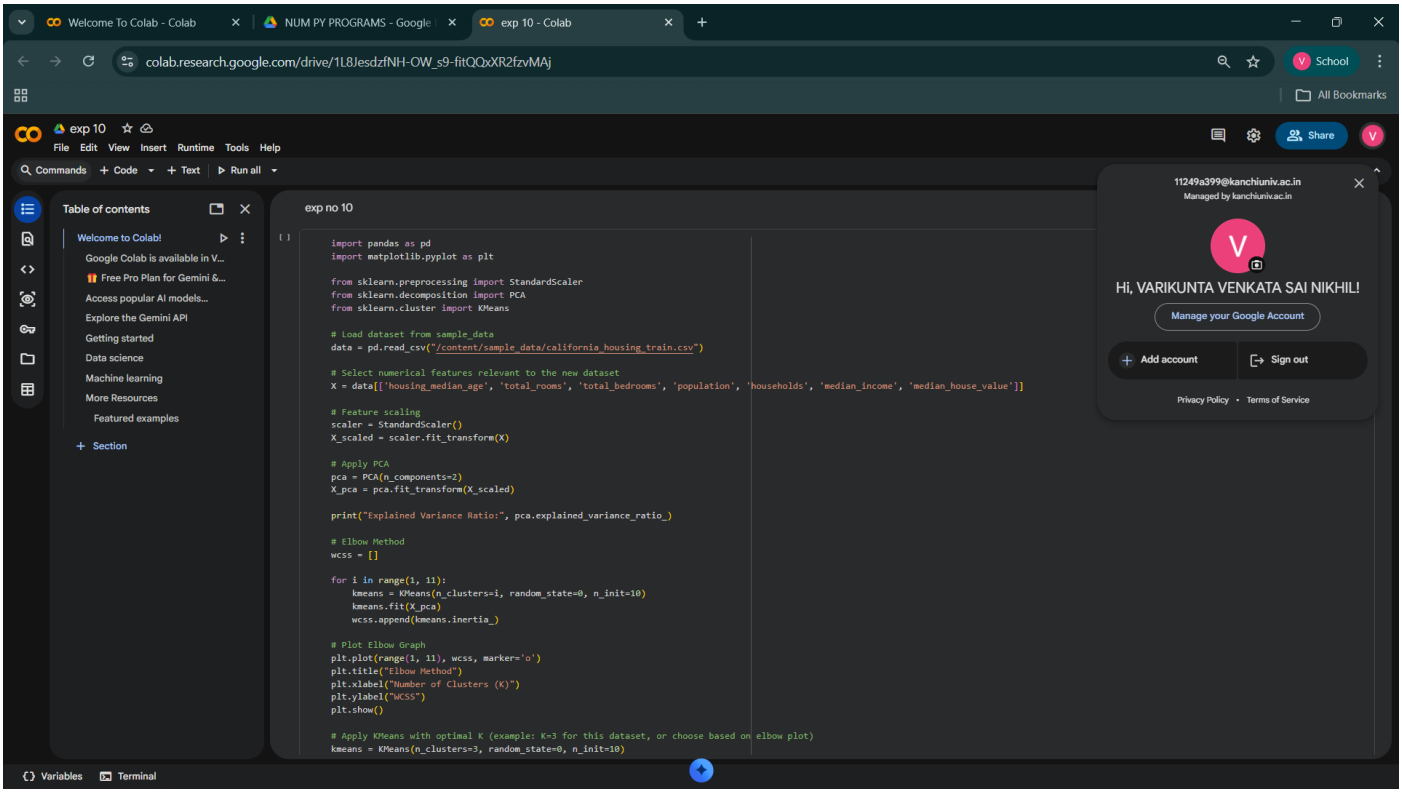
K-MEAN CLUSTERING WITH PCA AND ELBOW METHOD

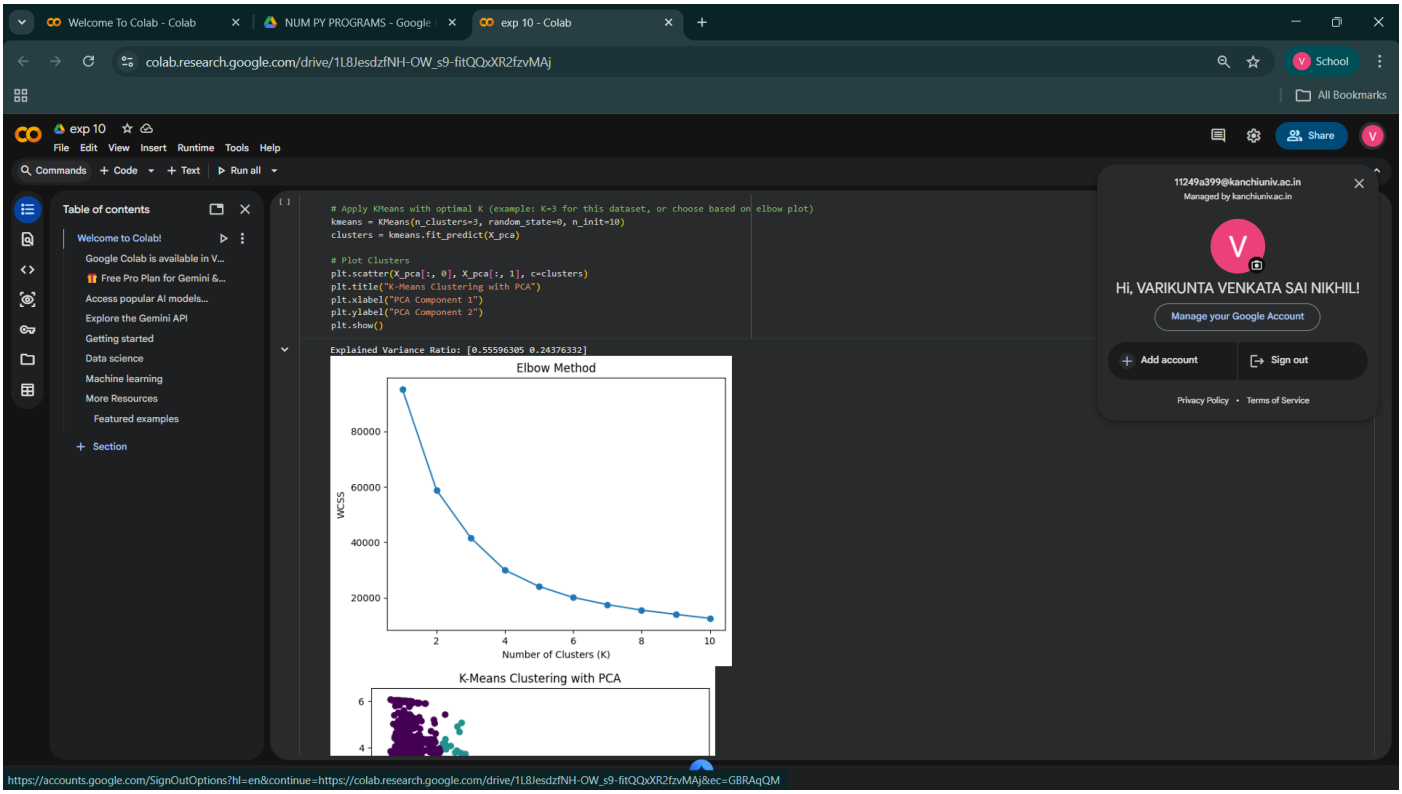
EXP-10

DATE-30-03-2026

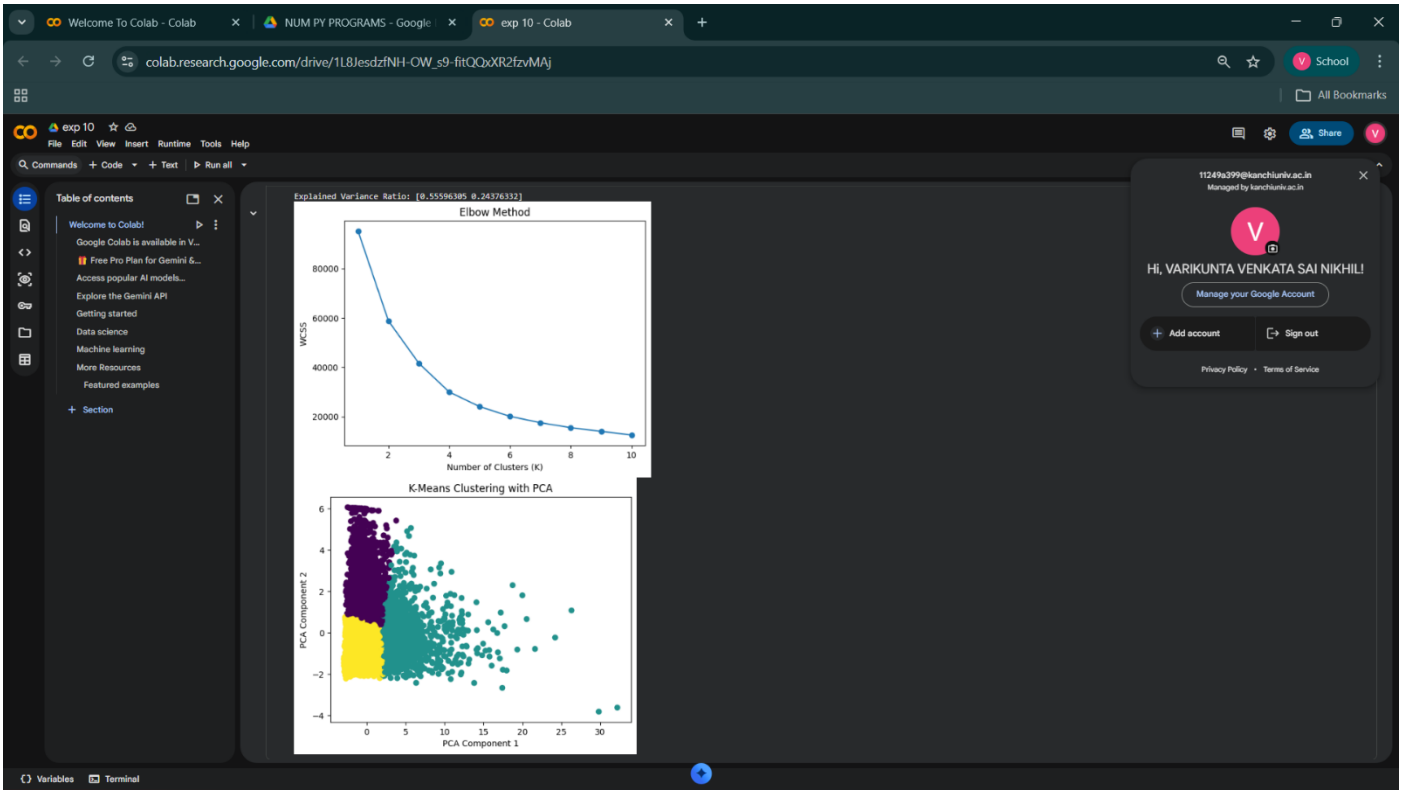
Aim

To perform K-Means clustering using Python along with PCA and the Elbow method.





OUTPUT



Result

K-Means clustering was successfully implemented. The optimal number of clusters was identified using the Elbow method, and PCA was used to reduce dimensionality for better visualization. The data was grouped into meaningful clusters effectively.